SPEAR3 RF Station

Wiring and Interconnection Reference

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This document records how the SPEAR3 RF station is wired: where each unit stands, the links between the buildings, every drawing in the repository with its status and its known errors, the connector pin tables, and the field checks still needed. The other documents describe how the station works (Doc L), what the new hardware must do (Doc C) and how the change is made (Doc M); the wiring they rely on is kept here, with the evidence for each line. It has two parts: the legacy station as it stands (I0), and the upgraded station, filled in at each step of the migration. All tables are generated from data files checked by a script; the data, not the tables, are edited.

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Revision History

Revision	Date	Scope of Change	Summary
R0	2026-11-11	Draft for review	Initial issue. The legacy station: its units by building and rack, the links between the buildings, a register of the 103 drawings in the repository, each read and given a final status, with what each shows and the documents that cite it, 22 errata of the drawings, the pin tables of the RF Local Display Panel's J2 and J3, of the two cables between the Fast Interlock Chassis and the AIM, of the panel's eleven fibers and of the HVPS junction box in B118, the 272 input and output points of the RF MPS PLC-5, and 24 field checks that gather the open wiring items of Docs 0, C, I, L and M. The upgraded station: its units by integration step.

1 Purpose and Scope

The SPEAR3 RF station spreads over four places: the HVPS power sections and their 12.47 kV contactors in B514, the HVPS controllers in B118, the control racks, klystron and waveguide in Room 101 of B132, and the cavities in the tunnel. Its wiring is drawn on about a hundred original drawings, some of which disagree with each other or with the hardware, and the other documents of the upgrade each describe the part they need. This document gathers it in one place:

- the units of the station and where they stand (§3);
- the links between the buildings and to the outside (Table 4);
- every drawing in the repository, with what is known of it (§4);
- the errors found in the drawings (§5);
- connector pin tables, each line with its verification level (§6);
- the checks on the hardware still needed (§7).

It records the wiring, not how the station works, which is in Doc L, nor what the new hardware must do, which is in Doc C, nor the order of the changes, which is in Doc M. Where another document lists the same pins it must agree with the tables here, and this document keeps the evidence for each line.

The legacy station, as it stands at I0, is described in §§3–7; the upgraded station in §8, which is filled in at each step of the migration (Doc M): its rule of configuration control asks that every hybrid configuration be recorded, and the connections are recorded here.

Two scales grade what is known. A drawing has a *status* (Table 1), which says how far it has been read and checked; a connection has a *verification level* (Table 2), which says what confirms it.

Table 1: Status of a drawing.

Status	Meaning
verified	Read and compared with other drawings, the PLC program, the EPICS database, photographs or measurements of the hardware, or an account of the installed system; no error found.
errata	Read; it has errors, listed in Table 7.
disputed	Read; it disagrees with another drawing, and which is right is not yet known (Table 7).
cited	Read for an item cited in the documents, but not compared with other sources.
read	Read: title and content recorded, not yet compared with other sources.

Status	Meaning
reference	Holds no wiring to check: a mechanical or fabrication drawing, a board layout, vendor data, or a design not installed in the station.
PEP-II	Drawn or written for PEP-II: background for SPEAR3, not a record of the station; where it is known to differ, the notes say so.
not read	In the repository but not yet read; its title is filled in when it is.

Table 2: Verification level of a connection.

Level	Name	Meaning
V0	Drawing only	Taken from one drawing, or from drawings that disagree.
V1	Sources agree	Two independent sources agree, such as two drawings, or a drawing and the PLC program; or their difference is explained by an erratum.
V2	Software confirms	The behavior seen in EPICS or in the running program confirms the line, for example a PV that follows a switch on it.
V3	Field verified	Checked on the hardware: continuity, a meter reading, or the label read on the cable.

2 How the Data Are Kept

Every table from §3 on is generated. The data are nine files of comma-separated values in the folder `data/` of this document (Appendix A). The build, `build.ps1 -Doc W`, first runs `scripts/docw-gen.js`, which reads the files, checks them and writes the tables into `generated/`; an error in the data stops the build. The script checks that:

- every identifier is unique: drawing number, erratum, field check, unit, connector contact, and PLC-5 module and point;
- every status is one defined here; the errata a drawing names exist and name the drawing in turn, and a drawing with the status *errata* or *disputed* names one;
- every revision keeps its leading zero;
- every file in the register exists, and the copies of one drawing are identical byte for byte, unless the notes record that they were compared page by page;
- every drawing file in the drawing folders of the repository is in the register, so that a new file is reported until it is entered;
- integration steps run from I0 to I8, and verification levels from V0 to V3.

It also finds where each drawing is cited, by reading the source of Docs 0, C, I, L, M, P and T for the drawing number, including lists such as “WD-340-330-21, -22 and -25”, and for the file name.

Rules for the data:

1. Edit the data, never the generated tables, and use a text editor: a spreadsheet program drops the leading zeros of the revisions when it saves the file.
2. Never reuse an identifier. A closed erratum or check keeps its number; its status changes and its result is recorded.
3. Raise a connection’s verification level only with the evidence recorded.
4. When a field check is done, record the result and the date in its row, update the rows it affects, and close the open items it names in the other documents at their next revision.

5. A note added to a drawing in the repository is recorded with its erratum.

3 The Legacy Station

Figure 1 shows the station by building and rack, Table 3 lists its units and Table 4 the links between the buildings and to the outside. The station has two HVPS units, SPEAR 1 and SPEAR 2, each with a power section in B514 and a controller in B118; one of them is connected at a time. The racks in B132 are numbered B132-11 to B132-15 from left to right, as in Doc L. The rack positions of the cross-connects X513, X517, X518, X519 and X522, between X530 and the PLC-5 modules and the HVPS lines, are not yet recorded.

Table 3: Units of the legacy station by location.

Unit	Model or drawing	Status	In service
B132-11			
VXI crate: IOC B132-IOCRF, CLK, RFP, IQA 1–3 *the IOC moves to new hardware at I8*	Elma 13-slot	in service	until I8
Arc/Interlock Module *VXI slot 12*	SD-340-309-01	in service	until I8

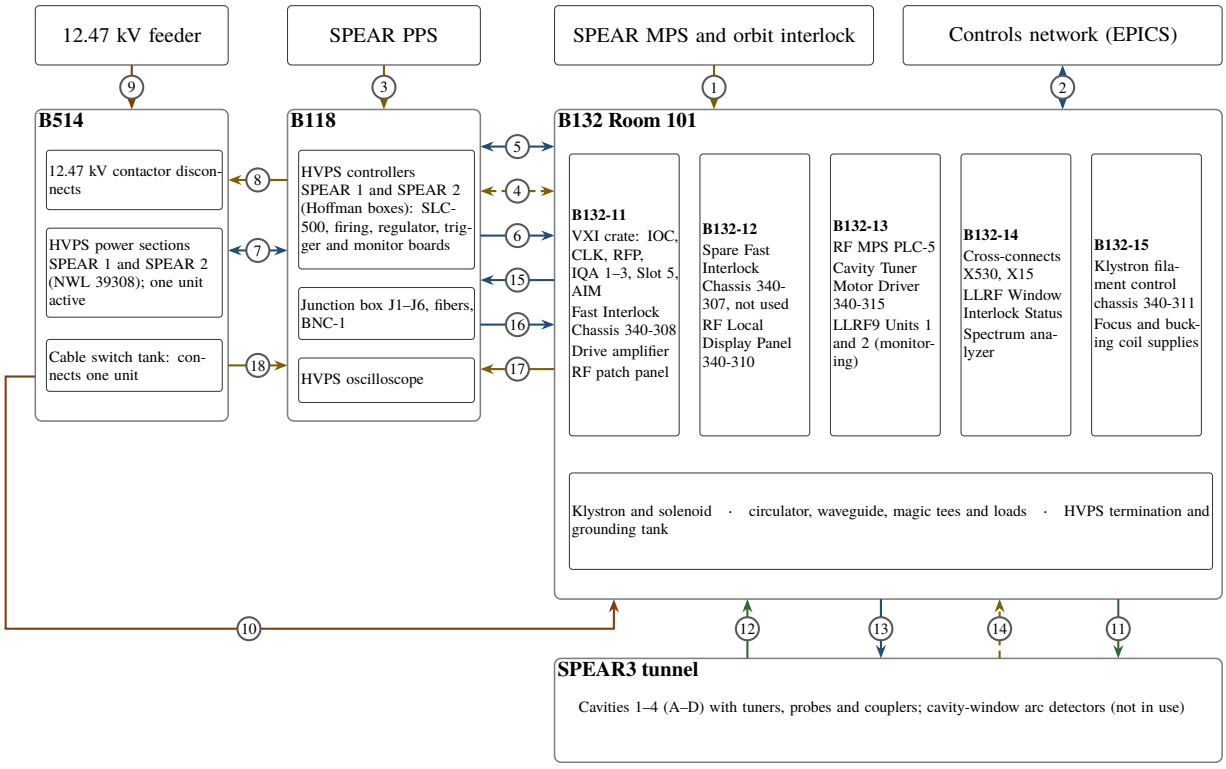


Figure 1: The legacy station (I0) by building and rack. Links are drawn to the building, not to the unit; the numbers refer to Table 4, which names the unit at each end. Brown: power; green: RF; blue: control and monitor signals; olive: interlocks and permits, dashed where on fiber. Arrows show the direction of the signals, double arrows both directions.

Unit	Model or drawing	Status	In service
MPS Shutoff module *VXI slot 5; SPEAR MPS and orbit permits in, J3 to the Fast Interlock Chassis spare-fault input*	drawing not found	in service	until I8
Fast Interlock Chassis *below the VXI crate; kept as fallback until I8*	340-308	in service	until I2
Drive amplifier	–	in service	–
RF patch panel	–	in service	–
B132-12			
Fast Interlock Chassis (spare) *above the panel; a slightly different version, not used*	340-307 SN04	spare	–
RF Local Display Panel *kept as fallback until I8*	340-310 SN12	in service	until I2
B132-13			
RF MPS PLC-5 (main 1771 crate and a second crate)	Allen-Bradley PLC-5	in service	until I4
Cavity Tuner Motor Driver: SLC-500 chassis with four 1746-HSTP1 modules *Remote I/O node 2; the motor cables move from the SS2000MD4 drivers to the Galil at I3, the chassis stays on the link and as fallback*	340-315 SN08	in service	until I8
LLRF9 Unit 1 *drives the station from I6*	Dimtel LLRF9	in service (monitoring)	–
LLRF9 Unit 2	Dimtel LLRF9	in service (monitoring)	–
B132-14			
Cross-connect X530 *panel cable J2 on terminals 1–25 and J3 on 26–50*	–	in service	–
Cross-connect X15 *waveguide pressure switches*	–	in service	–
LLRF Window Interlock Status chassis (2U) *above the spectrum analyzer; turns a legacy permit into an LLRF9 enable; replaced by the Interface Chassis at I2 (Doc C D8)*	built 2021	in service	until I2
Spectrum analyzer	–	in service	–
not recorded			
Cross-connect X519 (PLC-5 input module)	–	in service	–
Cross-connect X522 (PLC-5 output module)	–	in service	–
Cross-connect X517 (PLC-5 analog input module 2)	–	in service	–
Cross-connect X513 (lines from the HVPS controller)	–	in service	–
24 V bus X518 *feeds the output module and four panel spares*	–	in service	–
B132-15			
Klystron filament control chassis	340-311 SN14	in service	until I5
Main focus and bucking coil supplies	–	in service	–

Unit	Model or drawing	Status	In service
B132 Room 101			
Klystron and solenoid	Marconi	in service	–
Circulator, waveguide, magic tees and loads	–	in service	–
HVPS termination and grounding tank *Ross HV grounding switch, Danfysik DC-CT (cathode current), 3.3 mΩ return shunt, Pearson CT-110*	SD-730-790-05	in service	–
B118			
HVPS controllers SPEAR 1 and SPEAR 2 (Hoffman boxes) *SLC-500, Enerpro firing boards, regulator, trigger interconnect and monitor boards; one unit replaced at I7b, the other at I7c*	WD-730-790-02	in service	until I7
HVPS junction box (pluggable terminal blocks J1 to J6; fiber-optics box; BNC-1) *above the SPEAR 1 controller; connects the cables from B132 and B514 to the controller in service*	EI-444-636-00	in service	–
HVPS oscilloscope *channels: HVPS voltage, HVPS current, transformer current, SCR firing signal*	Tektronix	in service	–
B514			
HVPS power sections SPEAR 1 and SPEAR 2 *one unit active at a time*	NWL 39308	in service	–
HVPS cable switch tank *connects the −90 kV output of one unit to the cable to the termination tank*	EI-444-636-00	in service	–
12.47 kV contactor disconnects	GP-439-704-02	in service	–
SPEAR3 tunnel			
Cavities 1–4 (A–D) with tuners, probes and couplers	–	in service	–
Cavity-window arc detectors	–	not in use	–

Table 4: Links between buildings and to the outside (numbers as in the station overview).

No.	From	To	Medium	Content and drawings
1	SPEAR MPS and orbit interlock	VXI Slot 5 (B132-11)	wire	Two 24 V permits (J. Sebek, LLRF Interface Chassis specification); their wiring into the module is not recorded (FC2) *no drawing found*
2	Controls network	IOC B132-IOCRF (B132-11)	Ethernet	EPICS Channel Access
3	SPEAR PPS	HVPS controllers (B118)	wire	PPS 1 and PPS 2 permits *WD-730-790-02*

No.	From	To	Medium	Content and drawings
4	RF Local Display Panel (B132-12)	HVPS controllers (B118)	fiber	Three fibers: SCR ENABLE, KLYSTRON CROWBAR, STATUS; through the B118 junction box to the right- and left-side interface boards of the controller in service, re-patched at a switch-over *WD-340-330-29; SD-730-793-07; SD-730-793-08; EI-444-636-00*
5	VXI Remote I/O scanner (B132-11)	HVPS controllers (B118)	Remote I/O	SLC-500 adapter of each controller *WD-730-790-02*
6	HVPS monitor board (B118)	RF Local Display Panel (B132-12)	wire	HVPS voltage and current monitors (TS-3 of the controller in service, cables SPR02923 and SPR02924), through J5 of the B118 junction box, X513 and X530 *SD-730-793-12; WD-340-330-02; EI-444-636-00*
7	HVPS controllers (B118)	HVPS power sections (B514)	wire	SCR firing, 12 pairs to the phase-tank thyristor stacks; crowbar and voltage-divider cables *WD-730-790-01; WD-730-794-04*
8	HVPS controllers (B118)	Contactor disconnects (B514)	wire	TS-5, 15 terminals (Belden 83715): contactor on/off (MX coil), K-4 PPS control relay, RR remote reset relay *WD-730-790-01; WD-730-794-02*
9	12.47 kV feeder	Contactor disconnects (B514)	power	12.47 kV AC to the power sections *GP-439-704-02; ID-308-801-06*
10	HVPS power sections (B514)	Termination tank and klystron (B132)	HV cable	−90 kV DC: each unit's output to the cable switch tank in B514, which connects one unit; from it SPR02925 (−90 kV) and SPR02926 (ground) to the termination tank *EI-444-636-00; WD-730-790-01; SD-730-790-05*
11	Circulator and waveguide (B132)	Cavities (tunnel)	waveguide	RF power
12	Cavity probes and couplers (tunnel)	RF patch panel (B132-11)	coax	RF monitor signals *SD-340-330-01*
13	SS2000MD4 tuner motor drivers (B132)	Tuner motors (tunnel)	wire	Stepper motor cables
14	Arc detectors (tunnel)	Fast Interlock Chassis (B132-11)	fiber	Arc-detector fibers (cavity-window detectors not in use) *SD-340-308-02*

No.	From	To	Medium	Content and drawings
15	Termination tank (B132 Room 101)	HVPS controllers (B118)	wire and coax	Through the B118 junction box to the controller in service: SPR02944 and SPR02945 to TS-6 (terminals 9 to 14 and 1 to 6); SPR02946 for the oil-level switch LEV-3; SPR02947 coax from BNC-1 (klystron arc) to the left-side interface board. On TS-6: return-cable shunt on 20 and 21 to BNC-12, grounding-switch auxiliary contact on 11 and 12 (PPS chain 2 readback); Pearson CT-110 to J1 of SD-730-793-08. DC-CT output route not recorded *EI-444-636-00; WD-730-790-01; SD-730-790-05*
16	HVPS monitor boards (B118)	AIM (VXI slot 12; B132-11)	wire	9-conductor cable from the DB9 of the monitor board SD-730-793-12 of the controller in service, through J3 of the B118 junction box, to the AIM's DB9 *EI-444-636-00*
17	Pressure switch and PLC-5 (B132)	HVPS controllers (B118)	wire	Through J4 of the B118 junction box to TS-1 of the touch panel of the controller in service: a pressure switch labeled W6 PRESS, a flashing light, and PLC DC common from the PLC-5 DCM module *EI-444-636-00*
18	Flow switch beside the cable switch tank (B514)	HVPS controllers (B118)	wire	Two wires through J6 of the B118 junction box to TS-4 terminals 13 and 14 of the controller in service *EI-444-636-00*

4 Drawing Register

The register lists every drawing file in the drawing folders of the repository: `hvps/documentation/` (schematics, stack assemblies, switchgear and wiring diagrams), `llrf/documentation/` (coaxial cables, filament heater, the PEP-II block diagrams and system descriptions, the legacy interface modules, the RF Local Display Panel and the RF MPS wiring diagrams) and `pps/`, and the HVPS technical specification in `hvps/architecture/originalDocuments/`. It holds 103 drawings (Table 5): 45 verified against other sources, 20 with errata, 1 disputed, 19 reference documents with no wiring to check and 18 drawn or written for PEP-II. A drawing is verified against another drawing, the PLC program, the EPICS database, a photograph or measurement of the hardware, or an account of the installed system by the RF group (F. Wang's confirmation or J. Sebek's notes); the notes say which. Where the repository holds more than one copy of a drawing, all are listed. The last column of Table 6 gives the documents that cite the drawing, by number or by file name.

Table 5: Drawing status by area.

Area	Drawings	verified	errata	disputed	reference	PEP-II
HVPS	30	15	5	–	10	–
Switchgear	5	4	–	–	1	–
LLRF	1	–	–	–	–	1

Area	Drawings	verified	errata	disputed	reference	PEP-II
Heater	1	1	–	–	–	–
Architecture	14	–	–	–	–	14
Interface modules	3	3	–	–	–	–
Local panel	13	7	–	1	5	–
RF MPS	33	15	15	–	–	3
PPS	3	–	–	–	3	–
Total	103	45	20	1	19	18

Table 6: Drawing register.

Drawing	Title, notes and file	Status	Errata	Cited in
HVPS				
PS-341-360-01 rev 02	PEP-II RF system: 2.5 MW klystron power supply, technical specification *R. Cassel, Power Conversion Department, January 2004 (title page R2). A specification, not a drawing; no wiring. Doc 0 replaces its §1.5 phase-shifting transformer rating, 350 kVA, with the 2750 kVA nameplate of EI-730-790-00* `hvps/architecture/originalDocuments/ps3413600102.pdf`	reference		0, L, P
SD-237-230-12 rev 00	Enerpro general-purpose three-phase firing circuit, PCB FCOG6100 (Enerpro E128) *PP1 jumper block: phase reference (1–6) and gate-pulse profile (7–8). Agrees with J. Sebek's note on the installed board (R37 to R39 200 kΩ)* `hvps/documentation/schematics/sd2372301200.pdf`	verified		L
SD-237-230-14 rev 01	Analog regulator card *Compared with J. Sebek's regulator-board notes and the HVPS ladder program (N7:10 to output 0 of the NIO4V in slot 8)* `hvps/documentation/schematics/sd2372301401.pdf`	errata	E19	0, L
SD-730-790-01 rev 01	HVPS system schematic (top level) *Agrees with NWL's schematic EI-730-790-00, except that its input label rounds the 2750 kVA nameplate to 3000 kVA* `hvps/documentation/schematics/sd7307900101.pdf`	verified		0, L
SD-730-790-05 rev 01	Grounding tank schematic *Termination inductors 350 μH as built. J1 (MS3102R18-1P) agrees with WD-730-794-06* `hvps/documentation/schematics/sd7307900501.pdf` copy: `pps/sd7307900501.pdf`	verified		L
SD-730-791-03 rev 00	High-voltage SCR gate trigger board *Board circuit of the stack-mounted gate trigger board (1996); no inter-unit wiring* `hvps/documentation/stackAssemblies/StackDriver1sd73079103.pdf`	reference		–

Drawing	Title, notes and file	Status	Errata	Cited in
SD-730-793-03 rev 04	2 MW klystron PS 12 kV SCR driver board *Its 16 µs gate pulses and P1 rails (240 V, 120 V) agree with J. Sebek's HVPS hazards note* `hvps/documentation/schematics/sd7307930304.pdf`	verified		L
SD-730-793-04 rev 02	SCR crowbar trigger board (2002 version) *Supersedes SD-730-794-04. P1 as on SD-730-793-03 and -13* `hvps/documentation/schematics/sd7307930402.pdf`	verified		L
SD-730-793-07 rev 02	Right-side trigger interconnect board *SCR ENABLE fiber on J2, as on WD-340-330-29 and EI-444-636-00* `hvps/documentation/schematics/sd7307930702.pdf`	verified		L
SD-730-793-08 rev 01	Left-side trigger interconnect board *KLYSTRON CROWBAR and STATUS fibers on J2 and J3 and the klystron-arc BNC J1, as on WD-340-330-29 and EI-444-636-00* `hvps/documentation/schematics/sd7307930801.pdf`	verified		L
SD-730-793-12 rev 03	HVPS monitor board *Buffers the HVPS voltage and current sent to B132. Its DB9 agrees with EI-444-636-00, and its 10 kV/V scale with the divider WD-730-794-04 and the June 2020 readback* `hvps/documentation/schematics/sd7307931203.pdf`	verified		0, C, L
SD-730-793-13 rev 01	2 MW klystron PS optical SCR trigger board *Its 16 µs gate pulses agree with J. Sebek's HVPS hazards note; P1 as on SD-730-793-03* `hvps/documentation/schematics/sd7307931301.pdf`	verified		L
SD-730-794-04 rev 00	SCR crowbar trigger board (1999 version) *Superseded by SD-730-793-04: the same function and P1 connector* `hvps/documentation/schematics/sd7307940400.pdf`	verified		L
AD-730-792-01 rev 00	15 kV SCR assembly *Mechanical; no wiring; parts include PF-730-792-13 and -17* `hvps/documentation/stackAssemblies/ad7307920100.pdf`	reference		–
AD-730-792-08 rev 00	Crowbar tank: light crowbar assembly *Mechanical; no wiring* `hvps/documentation/stackAssemblies/ad7307920800.pdf`	reference		–
AD-730-794-16 rev 00	Klystron diode assembly, 50 kV *Mechanical; no wiring* `hvps/documentation/stackAssemblies/ad7307941600.pdf`	reference		–
PF-730-792-13 rev 00	15 kV AC SCR: cast SCR support *Mechanical; no wiring; part of AD-730-792-01* `hvps/documentation/stackAssemblies/pf7307921300.pdf`	reference		–
PF-730-792-17 rev 02	15 kV SCR: SCR end part *Mechanical; no wiring; part of AD-730-792-01 and -08 and AD-730-794-15 and -16* `hvps/documentation/stackAssemblies/pf7307921702.pdf`	reference		–
PF-730-792-22 rev 00	15 kV AC SCR crowbar: tension rod, 25 in *Mechanical; no wiring; part of AD-730-792-02* `hvps/documentation/stackAssemblies/pf7307922200.pdf`	reference		–

Drawing	Title, notes and file	Status	Errata	Cited in
PF-730-792-29 rev 00	Crowbar tank: tension rod, 20.75 in *Mechanical; no wiring; part of AD-730-792-01 and AD-730-794-16* hvps/documentation/stackAssemblies/pf7307922900.pdf	reference		–
PF-730-793-22 rev 01	15 kV AC light SCR: tension rod, 11.5 in *Mechanical; no wiring; part of AD-730-792-08* hvps/documentation/stackAssemblies/pf7307932201.pdf	reference		–
EI-444-636-00 rev 01	SPEAR3 2.5 MW klystron power supply system: electrical interconnect (3 sheets) *Matches the installed system (F. Wang, 11 November 2026). The source of links 4, 6, 10 and 15 to 18* hvps/documentation/wiringDiagrams/ei4446360001.pdf	errata	E16 E17 E18	–
EI-730-790-00 rev 00	Klystron power supply electrical schematic (NWL 39308) *The transformer manufacturer's schematic. Its switches agree with WD-730-794-05* hvps/documentation/wiringDiagrams/ei7307900000.pdf	verified		0, L
WD-730-790-01 rev 03	Interconnection wiring: B118 controller, contactor disconnect, termination tank hvps/documentation/wiringDiagrams/wd7307900103.pdf copy: pps/wd7307900103.pdf	errata	E2	L
WD-730-790-02 rev 06	Trigger enclosure wiring (B118 HVPS controller) *Master wiring diagram of the controller. Its I/O labels agree with the HVPS ladder program, and its SLC 5/03 processor in slot 0 is the one installed (F. Wang, 11 November 2026)* hvps/documentation/wiringDiagrams/wd7307900206.pdf copy: pps/wd7307900206.pdf	verified		0, L
WD-730-794-02 rev 00	SCR control driver, contactor interconnecting cables *TS-5 conductor list* hvps/documentation/wiringDiagrams/wd7307940200.pdf	errata	E3	L
WD-730-794-03 rev 00	SCR control driver: SCR interconnecting cables *Twelve SCR gate cables from the transformer tank (WD-730-792-01) to trigger boards H1 and H2, through P1 and P2 on 30 ft cables; thermocouples TC-1, TC-2 and oil level LEV1 to TS-6. Belden 88761 pairs as on WD-730-790-01* hvps/documentation/wiringDiagrams/wd7307940300.pdf	verified		–
WD-730-794-04 rev 00	SCR control driver, crowbar interconnecting cables *Includes the transformer tank (WD-730-792-01) and the output voltage dividers, whose ratio agrees with the monitor board SD-730-793-12 (10 kV/V) and the June 2020 readback* hvps/documentation/wiringDiagrams/wd7307940400.pdf	verified		0, L
WD-730-794-05 rev 03	SCR control driver: transformer cables *NWL termination box (pressure and vacuum, oil over-temperature, oil level, pressure relief, oil flow; CTs) to the Hoffman box; the same switches as EI-730-790-00* hvps/documentation/wiringDiagrams/wd7307940503.pdf	verified		–
WD-730-794-06 rev 00	Grounding tank wiring (test-stand configuration) *No revisions recorded* hvps/documentation/wiringDiagrams/wd7307940600.pdf copy: pps/wd7307940600.pdf	errata	E9	L

Switchgear

Drawing	Title, notes and file	Status	Errata	Cited in
GP-308-500-01 rev 03	12.47 kV vacuum contactor controller, schematic and general arrangement (LBL, 1977) *Authoritative relay legend; it names the relays of the as-built schematic GP-439-704-02 (MX, RR, TX)* `hvps/documentation/switchgear/gp3085000103.pdf`	verified		0, I, L
GP-439-704-02 rev 01	Vacuum contactor controller electrical schematic (as built, 2006) *Agrees with ID-308-801-06 and Ross 713203 on the PPS readback path (TB3-22 to 24)* `hvps/documentation/switchgear/gp4397040201.pdf` copy: `pps/gp4397040201.pdf`	verified		0, I, L
ID-308-801-06 rev 01	12.47 kV vacuum contactor controller, connection wiring (as built, 2006) *Agrees with GP-439-704-02 and Ross 713203 on the PPS readback path* `hvps/documentation/switchgear/id3088010601.pdf`	verified		0, I, L
Ross 713203 rev E	Vacuum contactor energy-storage closing and holding system (HCA-1-A driver, HQ3 contactor) *Vendor drawing. Agrees with ID-308-801-06 and GP-439-704-02 on the PPS readback path (auxiliary contact S5 on TB2-18 to 20)* `hvps/documentation/switchgear/rossEngr713203.pdf` copy: `pps/rossEngr713203.pdf`	verified		0, L
ABB DB 41-122M	ABB descriptive bulletin: Type MCO microprocessor time-overcurrent relay (devices 50/51), August 1996 *Vendor data, not a drawing; MCO relays are named on GP-439-704-02* `hvps/documentation/switchgear/DB41-122m MCO.pdf`	reference		–
LLRF				
SD-340-330-01 rev 00	Coax cable plant *PEP-II four-cavity coax cable plant (forward, reflected and probe of cavities A to D, klystron, circulator and loads); the SPEAR3 jacks are field check FC8* `llrf/documentation/coaxCables/sd3403300100.pdf`	PEP-II		L
Heater				
SD-340-311-00 rev 02	Klystron filament control chassis (340-311) *J1 agrees with WD-340-330-28 (filament on, motor up) and WD-340-330-21 (voltage and current monitors on terminals 11 and 12, channels 9 and 10)* `llrf/documentation/filamentHeater/sd3403110002.pdf`	verified		0, L, M
Architecture				
BD-340-330-00 rev 00	PEP-II LER RF station block diagram *PEP-II LER station LR44 with two cavities (P. Corredoura, January 1999); cited for the panel's beam-abort output* `llrf/documentation/legacyArchitecture/bd3403300000.pdf`	PEP-II		C, I, L, M
BD-340-330-01 rev 00	LLRF block diagram (alternative view) *PEP-II LER LLRF configuration (two cavities)* `llrf/documentation/legacyArchitecture/bd3403300100.pdf`	PEP-II		L

Drawing	Title, notes and file	Status	Errata	Cited in
BD-340-329-01 rev 00	PEP-II HER LLRF configuration, block diagram *PEP-II HER LLRF configuration (four cavities), February 1999; includes loops not installed at SPEAR3* `llrf/documentation/legacyArchitecture/blockDiagrambd3403290100-1.pdf`	PEP-II		–
PS-340-330-51 rev 00	PEP-II RF system description (Schwarz) *PEP-II system description; background in Docs L and P* `llrf/documentation/legacyArchitecture/ps3403305100.pdf`	PEP-II		L, P
PS-340-330-52 rev 00	LLRF feedback loop description (Schwarz) *PEP-II feedback loops; background in Docs L and P* `llrf/documentation/legacyArchitecture/feedbackLoopDescriptionps3403305200.pdf`	PEP-II		L, P
PS-340-330-53 rev 00	RF cavity low-power calibration procedure *PEP-II procedure (H. Schwarz), July 1999* `llrf/documentation/legacyArchitecture/ps3403305300.pdf`	PEP-II		–
PS-340-330-54 rev 00	RF station safety certification check-off list *PEP-II procedure, July 1999* `llrf/documentation/legacyArchitecture/ps3403305400.pdf`	PEP-II		–
PS-340-330-55 rev 03	RF station safety survey *PEP-II, revised December 2005* `llrf/documentation/legacyArchitecture/ps3403305503.pdf`	PEP-II		–
PS-340-330-56 rev 00	RF station coupling and cable calibration procedure *PEP-II procedure, July 1999* `llrf/documentation/legacyArchitecture/ps3403305600.pdf`	PEP-II		–
PS-340-330-57 rev 00	RF station full-power test and survey *PEP-II procedure, July 1999* `llrf/documentation/legacyArchitecture/ps3403305700.pdf`	PEP-II		–
PS-340-330-58 rev 00	RF station cavity phasing procedure *PEP-II procedure, July 1999* `llrf/documentation/legacyArchitecture/ps3403305800.pdf`	PEP-II		–
PS-340-330-59 rev 00	RF station turn-on procedure *PEP-II procedure, July 1999* `llrf/documentation/legacyArchitecture/ps3403305900.pdf`	PEP-II		–
PS-340-330-60 rev 01	Bellow cavity phasing procedure *PEP-II procedure, September 1999* `llrf/documentation/legacyArchitecture/ps3403306001.pdf`	PEP-II		–
PS-340-330-61 rev 02	RF non-ionizing radiation safety procedure *PEP-II procedure, January 2000* `llrf/documentation/legacyArchitecture/ps3403306102.pdf`	PEP-II		–
Interface modules				
SD-340-308-01 rev R1	Fast Interlock Chassis, transmitter board (command fibers) *Chassis 340-308 in service; its fiber jacks J45 to J50 agree with the rear photograph. The scan cuts off the number field: number and revision are from the file name* `llrf/documentation/legacyInterfaceModules/SD-340-308-01-R1-1of1.pdf`	verified		C, I, L, M

Drawing	Title, notes and file	Status	Errata	Cited in
SD-340-308-02 rev R1	Fast Interlock Chassis, receiver board (J15, J2, detector) *J15 and J2 agree pin for pin with the AIM schematic SD-340-309-01* `llrf/documentation/legacyInterfaceModules/SD-340-308-02-R1.pdf`	verified		C, I, L, M
SD-340-309-01 rev 02	Arc/Interlock Module (AIM) *2018 redraw. J3 and J4 agree pin for pin with SD-340-308-02; the enable word read in EPICS (0x71FB, 11 November 2026) fits its jumper logic on sheet 14* `llrf/documentation/legacyInterfaceModules/sd3403090102.pdf`	verified		0, C, I, L, M
Local panel				
AD-340-310-00 rev 00	RF Local Display Panel assembly, with the front-panel lamp layout *Lamp layout agrees with the legend SI-340-310-05 and the panel in service (340-310 SN12, photograph)* `llrf/documentation/localPanel/ad3403100000.pdf`	verified		–
AD-340-310-05 rev 00	RF Local Display Panel assembly, -05 version *Lamp layout agrees with the legend SI-340-310-05 and the panel in service (photograph)* `llrf/documentation/localPanel/ad3403100500.pdf`	verified		–
DL-340-310-00 rev 00	RF Local Display Panel drawing list *Lists twelve drawings of the panel set; all twelve are in the repository* `llrf/documentation/localPanel/dl3403100000.pdf`	verified		I, L, M
GP-340-310-01 rev 00	RF Local Display Panel PC card layout *PC card layout (lamp positions D2 to D44); no inter-unit wiring* `llrf/documentation/localPanel/gp3403100100.pdf`	reference		–
ML-340-310-00 rev 00	RF Local Display Panel material list *J2 female and J3 male (page 2), as on SD-340-311-01* `llrf/documentation/localPanel/ml3403100000.pdf`	verified		–
PC-340-310-01 rev 00	RF Local Display Panel PC card dimensions *PC card dimensions; no wiring* `llrf/documentation/localPanel/pc3403100100.pdf`	reference		–
PF-340-310-01 rev 00	RF Local Display Panel front panel, 15 3/4 in (fabrication) *Fabrication; no wiring* `llrf/documentation/localPanel/pf3403100100.pdf`	reference		–
PF-340-310-02 rev 00	RF Local Display Panel rear panel (fabrication) *Fabrication; connector cut-outs per SC-125-050-21* `llrf/documentation/localPanel/pf3403100200.pdf`	reference		–
PF-340-310-05 rev 02	RF Local Display Panel front panel (fabrication), revised *Fabrication; revisions added a hole and a dimension* `llrf/documentation/localPanel/pf3403100502.pdf`	reference		–
SD-340-311-01 rev 00	RF Local Display Panel schematic (PEP-II RF System Local Panel) *J2 AMP 745078-3 (female); J3 AMP 745078-1 (male)* `llrf/documentation/localPanel/sd3403110100.pdf`	disputed	E4	C, I, L, M

Drawing	Title, notes and file	Status	Errata	Cited in
SI-340-310-01 rev 00	RF Local Display Panel front-panel legend *Legend agrees with SI-340-310-05 and the panel in service (photograph)* `llrf/documentation/localPanel/si3403100100.pdf`	verified		–
SI-340-310-02 rev 00	RF Local Display Panel rear panel *Matches the rear photograph* `llrf/documentation/localPanel/si3403100200.pdf`	verified		I, L, M
SI-340-310-05 rev 01	RF Local Display Panel front panel *Legend agrees label for label with the panel in service (340-310 SN12, photograph)* `llrf/documentation/localPanel/si3403100501.pdf`	verified		I, L, M
RF MPS				
WD-340-330-02 rev 00	Wiring to the RF Local Display Panel (cross-connect X530) `llrf/documentation/mpsWiringDiagrams/wd3403300200.pdf`	errata	E4 E5 E7	C, I, L, M
WD-340-330-03 rev 00	Cavity junction boxes A to D: tuner limit switches, flow switches, position pots, IR detectors, tuner motor phases *J. Sebek's LLRF notes ask for its A-B module labels to be corrected* `llrf/documentation/mpsWiringDiagrams/wd3403300300.pdf`	errata	E10	–
WD-340-330-04 rev 00	Cavity vacuum: gauge controllers and ion-pump supplies of cavities A to D *Inputs agree with the PLC-5 program (I:032/1, /4, /11, /14 and I:033/1, /4, /11, /14)* `llrf/documentation/mpsWiringDiagrams/wd3403300400.pdf`	verified		–
WD-340-330-05 rev 00	Waveguide air (NIRP) interlock *Does not match SPEAR3 (J. Sebek)* `llrf/documentation/mpsWiringDiagrams/wd3403300500.pdf`	PEP-II		–
WD-340-330-06 rev 01	Waveguide air (NIRP) interlock *Does not match SPEAR3 (J. Sebek)* `llrf/documentation/mpsWiringDiagrams/wd3403300601.pdf`	PEP-II		–
WD-340-330-07 rev 00	PLC-5 thermocouple module 1 (1771-IXE): klystron collector, body, water and window air *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions), DCM table 2 words 4 to 11* `llrf/documentation/mpsWiringDiagrams/wd3403300700.pdf`	verified		–
WD-340-330-08 rev 00	Thermocouple module 2: return water of the waveguide loads and the circulator *Other channels agree with EPICS: crate power supply, ambient air, RF (drive) amplifier, circulator and three load HCW returns* `llrf/documentation/mpsWiringDiagrams/wd3403300800.pdf`	errata	E14	–
WD-340-330-09 rev 00	Thermocouple module 3: cavity A body 1 to 8 *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403300900.pdf`	verified		–
WD-340-330-10 rev 00	Thermocouple module 4: cavity A body 9 to 14 and tuners *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301000.pdf`	verified		–

Drawing	Title, notes and file	Status	Errata	Cited in
WD-340-330-11 rev 00	Thermocouple module 5: cavity A HOM loads, water, tuner bellows and spare *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions), except channel 8 (E15)* `llrf/documentation/mpsWiringDiagrams/wd3403301100.pdf`	errata	E15	–
WD-340-330-12 rev 00	Thermocouple module 6: cavity B body 1 to 8 *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301200.pdf`	verified		–
WD-340-330-13 rev 00	Thermocouple module 7: cavity B body 9 to 14 and tuners *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301300.pdf`	verified		–
WD-340-330-14 rev 00	Thermocouple module 8: cavity B HOM loads, tuner bellows and spare *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions), except channel 8 (E15)* `llrf/documentation/mpsWiringDiagrams/wd3403301400.pdf`	errata	E15	–
WD-340-330-15 rev 00	Thermocouple module 9: cavity C body 1 to 8 *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301500.pdf`	verified		–
WD-340-330-16 rev 00	Thermocouple module 10: cavity C body 9 to 14 and tuner *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301600.pdf`	verified		–
WD-340-330-17 rev 00	Thermocouple module 11: cavity C HOM loads, water, tuner bellows and spare *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions), except channel 8 (E15); the module is labeled MODULE 11 (E20)* `llrf/documentation/mpsWiringDiagrams/wd3403301700.pdf`	errata	E15 E20	–
WD-340-330-18 rev 00	Thermocouple module 12: cavity D body 1 to 8 *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions)* `llrf/documentation/mpsWiringDiagrams/wd3403301800.pdf`	verified		–
WD-340-330-19 rev 00	Thermocouple module 13: cavity D body 9 to 14 and tuner *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions); the module is labeled MODULE 25 (E20)* `llrf/documentation/mpsWiringDiagrams/wd3403301900.pdf`	errata	E20	–
WD-340-330-20 rev 00	Thermocouple module 14: cavity D HOM loads, tuner bellows and spare *Channel names agree with the legacy EPICS database (rf_temp_All and rf_temp_4CV substitutions), except channel 8 (E15)* `llrf/documentation/mpsWiringDiagrams/wd3403302000.pdf`	errata	E15	–

Drawing	Title, notes and file	Status	Errata	Cited in
WD-340-330-21 rev 00	PLC-5 analog input module 1 *Filament voltage and current on terminals 11 and 12 (channels 9 and 10), as on the filament chassis SD-340-311-00 and WD-340-330-28; terminals 13, 14 and 16 agree with WD-340-330-32; the channel order agrees with the EPICS database* `llrf/documentation/mpsWiringDiagrams/wd3403302100.pdf`	verified		C, M
WD-340-330-22 rev 00	PLC-5 analog input module 2 *The channel order agrees with the EPICS database; RF power level on terminal 12 (channel 10)* `llrf/documentation/mpsWiringDiagrams/wd3403302200.pdf`	errata	E6	C, M
WD-340-330-23 rev 00	PLC-5 digital input module 1 (1771-IBN, rack X4 module 9) `llrf/documentation/mpsWiringDiagrams/wd3403302300.pdf`	errata	E6 E7 E21 E22	C, I, M
WD-340-330-24 rev 00	PLC-5 digital input module 2 (2 sheets): cavity vacuum and flow interlocks *Cavity flow, vacuum and ion-pump current inputs (through X520) agree with the program (I:032/0, /1, /4, /10, /11, /14 and I:033/0, /1, /4, /10, /11, /14) and with WD-340-330-03 and -04* `llrf/documentation/mpsWiringDiagrams/wd3403302400.pdf`	verified		–
WD-340-330-25 rev 00	PLC-5 digital output module 1 (1771-OBN, rack X4 module 10) `llrf/documentation/mpsWiringDiagrams/wd3403302500.pdf`	errata	E6 E8	C, I, M
WD-340-330-26 rev 00	PLC-5 digital output module 2 (1771-OBN, rack X4 module 12), 2 sheets: all outputs drawn spare `llrf/documentation/mpsWiringDiagrams/wd3403302600.pdf`	errata	E11	–
WD-340-330-27 rev 00	Cavity tuner motor control: chassis 340-315 (1746-HSTP1) to the SLO-SYN M093-FC11 motors *Agrees with WD-340-330-03: 1746-HSTP1 slots X412-S1 to S4, X524, X521, motor driver chassis X404-J2 to J5, X507 to X510 and the cavity junction boxes* `llrf/documentation/mpsWiringDiagrams/wd3403302700.pdf`	verified		–
WD-340-330-28 rev 00	Klystron filament control: PLC-5 to the filament chassis 340-311 `llrf/documentation/mpsWiringDiagrams/wd3403302800.pdf`	errata	E12	–
WD-340-330-29 rev 00	Control fiber-optic cables of the RF Local Display Panel `llrf/documentation/mpsWiringDiagrams/wd3403302900.pdf`	errata	E1	C, I, L, M
WD-340-330-30 rev 00	Arc-detector fiber-optic cables (HER layout) *Combines the air- and vacuum-side detectors of each cavity on one channel (CH4 to CH7) through fiber coupling balls; the SPEAR3 database names separate air and vacuum channels* `llrf/documentation/mpsWiringDiagrams/wd3403303000.pdf`	PEP-II		–
WD-340-330-31 rev 00	Water-flow interlocks: klystron, circulator and cavities *Klystron and circulator inputs agree with the program (I:030/3 to /7, /10, /16)* `llrf/documentation/mpsWiringDiagrams/wd3403303100.pdf`	errata	E13	–
WD-340-330-32 rev 00	Klystron focus power supplies: main and bucking coil supplies to the PLC-5 *Agrees with WD-340-330-21 (terminals 13, 14 and 16) and WD-340-330-02 (X522-23, -24)* `llrf/documentation/mpsWiringDiagrams/wd3403303200.pdf`	verified		–

Drawing	Title, notes and file	Status	Errata	Cited in
WD-340-330-33 rev 00	Klystron window air: blower flow meter and blower supply *Flow input I:030/12 and blower output X522-33 (O:031/11) agree with the program* `llrf/documentation/mpsWiringDiagrams/wd3403303300.pdf`	verified		–
WD-340-330-34 rev 00	RF circulator controller: water relays, current and temperatures *Water-in and water-delta inputs agree with the program (I:030/13, /14)* `llrf/documentation/mpsWiringDiagrams/wd3403303400.pdf`	verified		L
PPS				
AD-137-181-00 rev 01	Modulator 6575 PPS interlock interface II: box assembly *Modulator 6575 design, not installed in the SPEAR3 station; Docs 0 and M adopt it for the PPS Interface Box (I7)* `pps/AD-137-181-00-01.pdf`	reference		–
SD-137-181-01 rev 02	Modulator 6575 PPS Interface II box: PC board (four 4PDT relays, channel A and B inhibit) *Modulator 6575 design (B. Morris), not installed in the SPEAR3 station; reference for the I7 PPS Interface Box* `pps/SD-137-181-01-02.pdf`	reference		–
WD-137-181-01 rev 03	PPS Interface II wiring diagram (modulator contactor) *Modulator 6575 design, not installed in the SPEAR3 station; revised for the Siemens PPS interface. File name ends -03-01* `pps/WD-137-181-01-03-01.pdf`	reference		–

5 Errata of the Drawings

An erratum is an error found in an original drawing, or a disagreement between two drawings that is not yet settled. 22 are recorded, 5 of them open (Table 7). The status *resolved* means the right reading is known; *documented*, that the error is known and harmless once recorded; *open*, that a field check is needed. Where the repository copy of a drawing carries a note, the status column says so; “here only” means that no other document records the erratum.

Table 7: Errata of the original drawings.

ID	Drawings	Finding and evidence	Status	Where
E1	WD-340-330-29	The Fast Interlock Chassis ends of the U13 and U14 fibers are labeled the wrong way round (FILAMENT and SOLENOID): the panel drives the solenoid request from U13 and the filament request from U14. *Panel schematic SD-340-311-01; legacy message logger and software. The baseline test of the panel (Doc M I0-T1) checks it.*	resolved *annotated*	Doc C; Doc L; Doc M
E2	WD-730-790-01	Coil labels swapped: RR is marked PPS and K-4 RESET. K-4 is the PPS control relay and RR the remote reset relay. *J. Sebek, Hoffman box PPS wiring note; relay legend of GP-308-500-01*	resolved *annotated (both copies)*	Doc L
E3	WD-730-794-02	Same RR and K-4 label swap as E2. *As E2*	resolved *annotated*	Doc L

ID	Drawings	Finding and evidence	Status	Where
E4	SD-340-311-01 WD-340-330-02	J3 numbered as mirror images of each other: the schematic's contact s is WD-340-330-02's contact 14 − s on the row of 13 contacts and 39 − s on the row of 12. Which numbering is stamped on the connector is not known. *J. Sebek, mapping spreadsheet (2022) and fiber note Rev 3*	open	Doc C O5; Doc M O2; Doc I O2; FC1
E5	WD-340-330-02	The J3 summary lines are labeled A-B DIGITAL INPUT although they come from the output module (X420-S10). *WD-340-330-25; J. Sebek, LLRF notes R2*	documented	here only
E6	WD-340-330-22 WD-340-330-23 WD-340-330-25	Panel side wrong: the panel is called X330 with connectors J1 and J2 (its J1 is the AC inlet); WD-23 puts digital inputs on X530-1 to -4, which WD-22 and WD-02 give to the HVPS voltage and current; WD-25 sheet 2 uses panel contacts 40 to 48 and X530 terminals 66 to 75. WD-23 also puts the reset input on X530-11, which WD-02 gives to HV DETECTED. *WD-340-330-02; panel rear photo*	documented	here only
E7	WD-340-330-23 WD-340-330-02	X519 numbered differently for the same module terminals (61, 62, 2, 3, 4 and 8 against 1, 2, 22, 23, 24 and 28). *Comparison of the two drawings; WD-340-330-33 numbers X519 as WD-02 does, by module terminal (X519 13 for terminal 13)*	open	here only; record as built at I4
E8	WD-340-330-25	Output O:030/17 marked SPARE although the program drives it (the waveguide-air MPS line to panel spare 1); the two coil-supply summaries O:031/12 and /13 stop at X522. *Decoded PLC-5 program; WD-340-330-02*	documented	here only
E9	WD-730-794-06	Drawn for the test stand: its manual ground-switch contact differs from the SPEAR3 grounding-tank schematic SD-730-790-05. *Doc L*	open	Doc L O1; FC11
E10	WD-340-330-03	Cavity B's position pot and IR detector are drawn to X420-S9-J1 06 and 09, a digital input module; through X516-06 and -09 they reach analog input module 1 (X430-S9), as cavity A's do (cavities C and D use analog module 2, X430-S10). *WD-340-330-21: cavity B tuner position on terminal 6 (channel 5), IR detector on terminal 9 (channel 8). X420-S9-J1 06 is the klystron body 2 flow switch on WD-340-330-31.*	documented	here only
E11	WD-340-330-26	Output 00 of output module 2 (X523-2) is marked SPARE, but the program drives it: O:032/0 follows the shield-rack air-temperature latch. What the output drives is not recorded. *Decoded PLC-5 program, rung 020, in every version from the original to that of February 2012*	open	FC22
E12	WD-340-330-28	Module labels X421-S10 and X431-S9 appear on no other sheet. The lines run through X522-25, -26 and X516-11, -12, which serve output module 1 (X420-S10) and analog input module 1 (X430-S9). *WD-340-330-21: filament voltage and current on terminals 11 and 12 (channels 9 and 10). WD-340-330-02: X522-25 is O:031/3, filament on.*	documented	here only
E13	WD-340-330-31	The four cavity flow switches are drawn to X420-S10-J1 01, 11, 21 and 31, the output module. WD-340-330-03 takes the same X520 terminals to input module 2 (X420-S11), and the program reads them as I:032/0, /10 and I:033/0, /10. *WD-340-330-03; decoded PLC-5 program*	documented	here only

ID	Drawings	Finding and evidence	Status	Where
E14	WD-340-330-08	Channel 3 of thermocouple module 2 is drawn SPARE, but it carries the HVPS oil temperature: EPICS reads it as SRF1:HVPSOIL:TEMP (DCM table 2, word 14), with alarms at 85 and 90 °C. *rf_hvps.db; the commented row in rf_temp_All.substitutions; read with caget on 11 November 2026: 60 °C, no alarm. F. Wang, 12 November 2026: the HVPS oil, not the magic-T post named in the 2014 log entry of rf_hvps.db*	documented	here only
E15	WD-340-330-11 WD-340-330-14 WD-340-330-17 WD-340-330-20	Channel 8 of thermocouple modules 5, 8, 11 and 14 is drawn SPARE, but it carries the ion-pump temperature of cavities A to D: EPICS reads SRF1:CAV1IONP:TEMP to CAV4IONP:TEMP. Only CAV3IONP has an alarm limit (HIGH 26 °C); the others have their limits at 401 °C, which disables them. *rf_temp_All and rf_temp_4CV substitutions; read with caget on 11 November 2026: 33, 46, 22 and 33 °C*	documented	here only
E16	EI-444-636-00	The two DB9 cables from junction-box strip J3 to the monitor boards SD-730-793-12 are both labeled SPEAR 1; the second is drawn among the SPEAR 2 destinations. *Sheet 1, B118 junction box*	documented	here only
E17	EI-444-636-00	The branch from junction-box strip J5 (LEV-3) to TS-6 terminals 7 and 8 of SPEAR 2 points to detail E on sheet 2, the SPEAR 1 sheet; every other SPEAR 2 branch points to sheet 3. *Sheet 1, B118 junction box; the title blocks of sheets 2 (SPEAR 1) and 3 (SPEAR 2)*	documented	here only
E18	EI-444-636-00	Sheets 2 and 3 label the thermocouple module in slot 3 and the input module in slot 7 of the controller PLC AB-1794-THERMC and AB-1794-IV16; 1794 is the catalog prefix of FLEX I/O, which does not fit an SLC 500 chassis. WD-730-790-02 labels them AB-1746-THERMC and AB-1746-IV16, beside an SLC 5/03 processor (1747-L532) in slot 0. *WD-730-790-02; the ladder program addresses slot 7 as a 1746-IV16; slot 0 holds an SLC 5/03 (F. Wang, 11 November 2026)*	open	FC23
E19	SD-237-230-14	Pin J4A (+EL1) is labeled POS. VOLTAGE LIMIT COMMAND and drawn as an output, but in the controller it is the voltage reference input, driven across J4-1 and J4-7 by output 0 of the NIO4V in PLC slot 8. *J. Sebek, regulator-board notes; the HVPS ladder program, which writes N7:10 to that output*	documented	Doc L
E20	WD-340-330-17 WD-340-330-19	The thermocouple modules of rack X420 in slots 3 and 5 are labeled MODULE 11 and MODULE 25. Their neighbours run 1, 2, 4 and 6, the program addresses them as rack 2, groups 2 and 4, and slot 11 of X420 holds digital input module 2. *Headers of WD-340-330-15 to -20 and -24; block transfers of the decoded PLC-5 program*	documented	here only

ID	Drawings	Finding and evidence	Status	Where
E21	WD-340-330-23	On sheet 1 the labels of bits 12 to 16 of I:030 are one row late. The far ends and the program put klystron window air flow on bit 12, circulator water input temperature on 13, circulator water delta temperature on 14, waveguide air pressure on 15 and circulator flow on 16; the drawing labels them CIRCULATOR WATER INPUT TEMP, CIRCULATOR WATER DELTA TEMP, WAVEGUIDE AIR PRESSURE, CIRCULATOR FLOW and LER/HER STATION TYPE. *Decoded PLC-5 program (KLY_AIR_FLOW, CIRC_WATER_IN, CIRC_DELTA, WG_AIR, CIRC_FLOW); WD-340-330-31, -33 and -34*	documented	here only
E22	WD-340-330-23	Input I:031/04 (terminal 25) is drawn SPARE and tied to +24 V through X518 1 and X519 5, but the program reads it as VXI_SUM_DI and latches NIRP_IN if it drops; the tie holds it on. *Decoded PLC-5 program, rungs DIGI_IN1:003 and 004*	documented	here only

6 Connector Tables

6.1 RF Local Display Panel J2 and J3

The RF Local Display Panel 340-310 in rack B132-12 meets the RF MPS PLC-5 through two 25-contact connectors on its rear, J2 and J3, cabled to cross-connect X530: J2 to terminals 1 to 25 and J3 to terminals 26 to 50. The panel schematic SD-340-311-01 gives J2 as female and J3 as male, as does the material list ML-340-310-00. The tables give for each contact the labels of the schematic and of the cross-connect diagram WD-340-330-02, the direction, the X530 terminal, the next point on the way to the PLC-5, the PLC-5 point, the use the Interface Chassis makes of the line (its identifiers P1 to P12 and N1 to N4, Doc C), and the verification level. The panel leaves service at I2, when the Interface Chassis takes over the lines it uses; the lines the panel only displays end there.

Table 8: RF Local Display Panel J2 (SYSTEM I/O, female), with the pin numbers of SD-340-311-01.

Pin	Schematic / WD-02 label	Direction	X530	Next	PLC-5	Chassis	V
1	Vin+ (from HVPS) / HVPS Vin+	from HVPS	1	X513-6	–	N1 +	V1
2	Iin+ (from HVPS) / HVPS Iin+	from HVPS	2	X513-11	–	N2 +	V1
3	Vout (to AB) / HVPS Vout	to PLC-5	3	X517-13	AI 2 ch 11	N3	V1
4	Iout (to AB) / HVPS Iout	to PLC-5	4	X517-14	AI 2 ch 12	N4	V1
5	GND / Return	return	5	return bus	–	return	V1
6	SolenoidM Ctrl24 / Main Focus On	to panel	6	X522-23	O:031/1	display only	V1
7	SolenoidM Fault15 / Main Focus PS Interlock	to panel	7	X522-34	O:031/12	P1	V1
8	GND / Return	return	8	return bus	–	return	V1
9	Filament Ctrl24 / Filament On	to panel	9	X522-25	O:031/3	display only	V1

Pin	Schematic / WD-02 label	Direction	X530	Next	PLC-5	Chassis	V
10	Timeout Ctrl24 / Filament Timer Bypass from A-B	to panel	10	X522-7	O:030/5	display only	V1
11	HV Detected24 / HV Detected	to panel	11	X522-29	O:031/7	display only	V1
12	Local Mode24 / Station Local/Remote Mode	to PLC-5	12	X519-2	I:030/1	P12 (open)	V1
13	GND / Return	return	13	return bus	–	return	V1
14	Vin- (from HVPS) / HVPS Vin-	from HVPS	14	X513-7	–	N1 −	V1
15	Iin- (from HVPS) / HVPS Iin-	from HVPS	15	X513-12	–	N2 −	V1
16	AGnd / Return	return	16	return bus	–	N3 return	V1
17	AGnd / Return	return	17	return bus	–	N4 return	V1
18	Solenoid Req24 / Online/Offline Request to A-B	to PLC-5	18	X519-24	I:031/3	P7	V1
19	SolenoidB Ctrl24 / Bucking Coil PS On/Off	to panel	19	X522-24	O:031/2	display only	V1
20	SolenoidB Fault15 / Bucking Coil PS Interlock	to panel	20	X522-35	O:031/13	P2	V1
21	Filament Req24 / Filament PS On/Off	to PLC-5	21	X519-22	I:031/1	P8	V1
22	Timeout Req24 / Filament Timer Bypass	to PLC-5	22	X519-23	I:031/2	P9	V1
23	Reset Req24 / Interlock Reset	to PLC-5	23	X519-1	I:030/0	P10	V1
24	RF detected24 / RF On Indicator	to panel	24	X522-32	O:031/10	display only	V1
25	Online mode24 / Online/Offline Control	to PLC-5	25	X519-28	I:031/7	P11 (held on)	V1

J2 is at level V1: the schematic and WD-340-330-02 give the same contact for every line. J3 stays at V0 (erratum E4). The two drawings number J3 as mirror images of each other: the schematic's contact s is WD-340-330-02's contact $14 - s$ on the row of 13 contacts and $39 - s$ on the row of 12. Both give the same order of signals; they differ only in the end from which the contacts are numbered. The first column of Table 9 follows the schematic and the second gives WD-340-330-02's number; the X530 terminal is 25 plus WD-340-330-02's number. Field check FC1 (Table 19) settles which numbering is stamped on the connector.

Table 9: RF Local Display Panel J3 (FROM A/B INPUTS, male): the contact of SD-340-311-01 and of WD-340-330-02 (erratum E4).

Pin	WD-02	Schematic / WD-02 label	Direction	X530	Next	PLC-5	Chassis	V
1	13	Cavity1 Summary24 / Cavity A Sum	to panel	38	X522-2	O:030/0	display only	V0
2	12	Cavity3 Summary24 / Cavity C Sum	to panel	37	X522-4	O:030/2	display only	V0
3	11	Klystron Summary24 / Klystron Sum	to panel	36	X522-6	O:030/4	display only	V0
4	10	Focus Summary24 / Focus Sum	to panel	35	X522-8	O:030/6	display only	V0

Pin	WD-02	Schematic / WD-02 label	Direction	X530	Next	PLC-5	Chassis	V
5	9	Magic Tee2 Summary24 / Magic Tee 2 Sum	to panel	34	X522-12	O:030/10	display only	V0
6	8	Waveguide Pressure24 / Waveguide Air	to panel	33	X522-14	O:030/12	P3	V0
7	7	Water Flow Summary24 / Flow Sum	to panel	32	X522-16	O:030/14	display only	V0
8	6	Temperature Summary24 / Temp Sum	to panel	31	X522-18	O:030/16	display only	V0
9	5	AB Summary24 / Interlock Sum	to panel	30	X522-22	O:031/0	P5	V0
10	4	Spare2 24 / Spare 2	+24 V	29	X518	–	open	V0
11	3	Spare4 24 / Spare 4	+24 V	28	X518	–	open	V0
12	2	GND / Return	return	27	return bus	–	return	V0
13	1	GND / Return	return	26	return bus	–	return	V0
14	25	Cavity2 Summary24 / Cavity B Sum	to panel	50	X522-3	O:030/1	display only	V0
15	24	Cavity4 Summary24 / Cavity D Sum	to panel	49	X522-5	O:030/3	display only	V0
16	23	Collector Summary24 / Collector Sum	to panel	48	X522-28	O:031/6	P4	V0
17	22	Magic Tee1 Summary24 / Magic Tee 1 Sum	to panel	47	X522-9	O:030/7	display only	V0
18	21	Magic Tee3 Summary24 / Magic Tee 3 Sum	to panel	46	X522-13	O:030/11	display only	V0
19	20	Circulator Summary24 / Circulator Sum	to panel	45	X522-15	O:030/13	display only	V0
20	19	Vacuum Summary24 / Vacuum Sum	to panel	44	X522-17	O:030/15	display only	V0
21	18	AB Heartbeat24 / A-B Heartbeat	to panel	43	X522-27	O:031/5	P6	V0
22	17	Spare1 24 / Waveguide Air MPS	to panel	42	X522-19	O:030/17	open	V0
23	16	Spare3 24 / Spare 3	+24 V	41	X518	–	open	V0
24	15	Spare5 24 / Spare 5	+24 V	40	X518	–	open	V0
25	14	(spare) / N/C	none	39	–	–	open	V0

Six contacts of J3 need a remark. Four, the panel's spares 2 to 5, carry 24 V from bus X518 at all times. Spare 1 is not spare: the program drives output O:030/17 onto it, the waveguide-air MPS line (erratum E8). Contact 25 is not connected.

6.2 Fast Interlock Chassis to AIM

Two cables join the Fast Interlock Chassis 340-308 to the AIM in VXI slot 12: a 25-pin cable from the chassis's J15 to the AIM's J3, which carries the six commands, the AB summary and heartbeat, the AIM's fused supplies and twelve trip-test lines, and a 37-pin cable from J2 of the chassis's receiver board to the AIM's J4, which carries the twelve arc signals and the spare-fault line. At I2 the Interface Chassis takes over the chassis end of both cables (Doc C, IC-24 and IC-27a); the AIM end stays until I8.

The tables are read from both ends: the Fast Interlock Chassis receiver board SD-340-308-02, a scan, and the AIM schematic SD-340-309-01, a CAD drawing whose wires were traced pin by pin from the drawing's own geometry. The two agree on every pin, so every line is at level V1. The chassis numbers its arc channels and trip tests from 1 and the AIM from 0: its PEAK_n is the AIM's ARC$\pm(n-1)$ and its TEST_CHn the AIM's TT$(n-1)$. On the 37-pin cable the receiver board grounds the negative wire of each arc pair, and it connects nothing to pins 17 to 19 and 35 to 37, where the AIM puts its ground and its fused $5\,\mathrm{V}$.

Table 10: Cable from J15 of the Fast Interlock Chassis (AMP 745592-4, SD-340-308-02 sheet 1) to J3 of the AIM (SD-340-309-01 sheet 18).

Pin	Signal	Fast IC end	AIM end	Sense	Interface Chassis	V
1	FILAMENT_ON *Fiber J45 to the panel; panel receiver U14 (E1)*	FILAMENT_ON	FILAMENT_ON (U45 Y1)	Low: request	V1 to P8	V1
14	SOLENOID_ON *Fiber J46 to the panel; panel receiver U13 (E1)*	SOLENOID_ON	SOLENOID_ON (U45 Y2)	Low: request	V2 to P7; online command of the logic	V1
2	HVPS_ON	HVPS_ON	HVPS_ON (U45 Y3)	Low: request	V3; HVPS-on command of the logic	V1
15	BEAM_ABORT *The AIM's net is ABORT_BAR before its inverting buffer*	BEAM_ABORT	ABORT (U45 Y4)	High: abort	not connected (Doc C IC-20)	V1
3	FILAMENT_ TIMEOUT	FILAMENT_ TIMEOUT	FILAMENT_ TIMEOUT (U45 Y5)	Low: request	V4 to P9	V1
16	FAULT_RESET	FAULT_RESET	FAULT_RESET (U45 Y6)	Low: reset	V5 to P10	V1
4	+12 V *Powers the Fast Interlock Chassis*	+12V_I	FUSED_+12V	–	not connected	V1
17	Ground	GND	AGND	–	reference	V1
5	-12 V *Powers the Fast Interlock Chassis*	-12V_I	FUSED_-12V	–	not connected	V1
18	Ground	GND	GND	–	reference	V1
6	+5 V *Powers the Fast Interlock Chassis*	+5V_I	FUSED_VDD	–	not connected	V1
19	Trip test 1	TEST_CH1	TT0	Low pulse	not used	V1
7	Trip test 2	TEST_CH2	TT1	Low pulse	not used	V1
20	Trip test 3	TEST_CH3	TT2	Low pulse	not used	V1
8	Trip test 4	TEST_CH4	TT3	Low pulse	not used	V1
21	Trip test 5	TEST_CH5	TT4	Low pulse	not used	V1
9	Trip test 6	TEST_CH6	TT5	Low pulse	not used	V1

Pin	Signal	Fast IC end	AIM end	Sense	Interface Chassis	V
22	Trip test 7	TEST_CH7	TT6	Low pulse	not used	V1
10	AB summary *From fiber J30*	AB_SUMMARY	AB_SUMMARY_ FLT (U42B)	Low: no fault	V6 from P5	V1
23	AB heartbeat *From fiber J29*	HEART_BEAT	HEARTBEAT (U42C)	Low during each pulse	V7 from P6	V1
11	Trip test 8	TEST_CH8	TT7	Low pulse	not used	V1
24	Trip test 9	TEST_CH9	TT8	Low pulse	not used	V1
12	Trip test 10	TEST_CH10	TT9	Low pulse	not used	V1
25	Trip test 11	TEST_CH11	TT10	Low pulse	not used	V1
13	Trip test 12	TEST_CH12	TT11	Low pulse	not used	V1

Table 11: 37-pin cable from J2 of the Fast Interlock Chassis receiver board (SD-340-308-02 sheet 1) to J4 of the AIM (SD-340-309-01 sheet 18).

Pin	Signal	Fast IC end	AIM end	Sense	Interface Chassis	V
20	Arc channel 1 +	PEAK_1	ARC+0	–	terminated at no arc (Doc C IC-27c)	V1
1	Arc channel 1 −	ground	ARC−0	–	terminated at no arc (Doc C IC-27c)	V1
21	Arc channel 8 +	PEAK_8	ARC+7	–	terminated at no arc (Doc C IC-27c)	V1
2	Arc channel 8 −	ground	ARC−7	–	terminated at no arc (Doc C IC-27c)	V1
22	Arc channel 2 +	PEAK_2	ARC+1	–	terminated at no arc (Doc C IC-27c)	V1
3	Arc channel 2 −	ground	ARC−1	–	terminated at no arc (Doc C IC-27c)	V1
23	Arc channel 9 +	PEAK_9	ARC+8	–	terminated at no arc (Doc C IC-27c)	V1
4	Arc channel 9 −	ground	ARC−8	–	terminated at no arc (Doc C IC-27c)	V1
24	Arc channel 3 +	PEAK_3	ARC+2	–	terminated at no arc (Doc C IC-27c)	V1
5	Arc channel 3 −	ground	ARC−2	–	terminated at no arc (Doc C IC-27c)	V1
25	Arc channel 10 +	PEAK_10	ARC+9	–	terminated at no arc (Doc C IC-27c)	V1
6	Arc channel 10 −	ground	ARC−9	–	terminated at no arc (Doc C IC-27c)	V1
26	Arc channel 4 +	PEAK_4	ARC+3	–	terminated at no arc (Doc C IC-27c)	V1
7	Arc channel 4 −	ground	ARC−3	–	terminated at no arc (Doc C IC-27c)	V1

Pin	Signal	Fast IC end	AIM end	Sense	Interface Chassis	V
27	Arc channel 11 +	PEAK_11	ARC+10	–	terminated at no arc (Doc C IC-27c)	V1
8	Arc channel 11 −	ground	ARC−10	–	terminated at no arc (Doc C IC-27c)	V1
28	Arc channel 5 +	PEAK_5	ARC+4	–	terminated at no arc (Doc C IC-27c)	V1
9	Arc channel 5 −	ground	ARC−4	–	terminated at no arc (Doc C IC-27c)	V1
29	Arc channel 12 +	PEAK_12	ARC+11	–	terminated at no arc (Doc C IC-27c)	V1
10	Arc channel 12 −	ground	ARC−11	–	terminated at no arc (Doc C IC-27c)	V1
30	Arc channel 6 +	PEAK_6	ARC+5	–	terminated at no arc (Doc C IC-27c)	V1
11	Arc channel 6 −	ground	ARC−5	–	terminated at no arc (Doc C IC-27c)	V1
32	Arc channel 7 +	PEAK_7	ARC+6	–	terminated at no arc (Doc C IC-27c)	V1
13	Arc channel 7 −	ground	ARC−6	–	terminated at no arc (Doc C IC-27c)	V1
34	Spare fault *The AIM's External MPS channel; today from Slot 5 J3 through the front LEMO*	SPARE_FAULT	SPARE_FLT (U42A; 10 kΩ pull-up)	Low: no fault	V8: low while S1 and S2 are present	V1
12	Ground	ground	AGND	–	reference	V1
14	Ground	ground	AGND	–	reference	V1
15	Ground	ground	AGND	–	reference	V1
16	Ground	ground	AGND	–	reference	V1
31	Ground	ground	AGND	–	reference	V1
33	Ground	ground	AGND	–	reference	V1
17	Ground	not connected	AGND	–	reference	V1
18	Ground	not connected	AGND	–	reference	V1
19	Ground	not connected	AGND	–	reference	V1
35	+5 V	not connected	FUSED_VDD (F4)	–	not connected	V1
36	+5 V	not connected	FUSED_VDD (F4)	–	not connected	V1
37	+5 V	not connected	FUSED_VDD (F4)	–	not connected	V1

6.3 Fibers of the RF Local Display Panel

Eleven fibers leave the panel (Table 12): eight to and from the Fast Interlock Chassis, which carry the AIM's commands and return the AB summary and heartbeat, and three to and from the HVPS controllers in B118. The panel devices and far ends are those of WD-340-330-29, whose labels at the Fast Interlock Chassis end

of the U13 and U14 fibers are swapped (erratum E1). The chassis jacks are read from its rear panel and, for J29 and J30, its receiver board; the HVPS jacks from the trigger interconnect boards SD-730-793-07 and -08. At I2 the eight Fast Interlock Chassis fibers become internal to the Interface Chassis, which drives the three HVPS fibers as F1 to F3 (Doc C).

Table 12: The eleven fibers of the RF Local Display Panel (WD-340-330-29, with erratum E1; Fast Interlock Chassis jacks from its rear panel; HVPS jacks from SD-730-793-07 and -08).

Fiber	Signal	Far end	Panel end	Light means	Interface Chassis	V
U10	VXI beam abort	Fast IC J48 (BEAM_ ABORT)	Receiver U10	Lit: no abort	not taken over (Doc C D4)	V1
U11	HVPS STATUS *Through the B118 junction box; re-patched to the unit in service at a switch-over*	Left-side board J3 (HFBR-1412 transmitter)	Receiver U11	Lit: controller enabled; no crowbar	F3	V1
U12	HVPS on request	Fast IC J47 (HVPS_ ON)	Receiver U12	Lit: request	internal from I2	V1
U13	Station online request *WD-340-330-29 labels this end FILAMENT (E1)*	Fast IC J46 (SOLENOID_ON)	Receiver U13	Lit: request	internal from I2	V1
U14	Filament and focus request *WD-340-330-29 labels this end SOLENOID (E1)*	Fast IC J45 (FILAMENT_ON)	Receiver U14	Lit: request	internal from I2	V1
U15	Filament timeout bypass	Fast IC J49 (FILAMENT_ TIMEOUT)	Receiver U15	Lit: bypass	internal from I2	V1
U16	Reset	Fast IC J50 (FAULT_ RESET)	Receiver U16	Lit: reset	internal from I2	V1
U19	HVPS triggers enable (SCR ENABLE) *Through the B118 junction box; re-patched to the unit in service at a switch-over*	Right-side board J2 (HFBR-2412 receiver)	Transmitter U19	Lit: triggers enabled	F1	V1
U20	HVPS crowbar (KLYSTRON CROWBAR) *Through the B118 junction box; re-patched to the unit in service at a switch-over*	Left-side board J2 (HFBR-2412 receiver)	Transmitter U20	Lit: no crowbar	F2	V1
U21	AB heartbeat	Fast IC J29 (HEARTBEAT)	Transmitter U21	Pulsing	internal from I2	V1

Fiber	Signal	Far end	Panel end	Light means	Interface Chassis	V
U22	AB summary	Fast IC J30 (A/B SUMMARY)	Transmitter U22	Lit: no fault	internal from I2	V1

6.4 HVPS Junction Box in B118

The cables from B132 and from the cable switch tank in B514 end on a junction box above the SPEAR 1 controller in B118 (Table 13). EI-444-636-00 draws each strip wired to both units; the switch-over procedure (J. Sebek, R0, January 2023, `hvps/documentation/procedures/spearRfHvpsSwitchProcedureR0.docx`), which calls the junction box the switchbox, shows that one unit is connected at a time: the strips J1 to J6 are 10-position pluggable terminal blocks which, with the three panel fibers and the klystron-arc BNC, are plugged into the controller in service. The table follows EI-444-636-00, which matches the installed system (F. Wang, 11 November 2026); errata E16 and E17 record two drafting slips on its sheet 1.

Table 13: The HVPS junction box in B118 (EI-444-636-00 sheet 1): its pluggable strips and BNC-1 connect the controller in service.

Strip	Lines	Far end in B132 or B514	End in the controller in service	V
J1	Termination tank: TS-6 lines (9 conductors)	SPR02944 from the termination tank	TS-6 terminals 9 to 14	V1
J2	Termination tank: TS-6 lines (9 conductors)	SPR02945 from the termination tank	TS-6 terminals 1 to 6	V1
J3	AIM DB9 (9 conductors) *The drawing labels both monitor boards SPEAR 1 (E16)*	AIM DB9 (VXI slot 12; B132-11)	DB9 of the monitor board SD-730-793-12	V1
J4 1–3	Pressure switch W6 PRESS	Pressure switch (B132)	TS-1 of the touch panel	V1
J4 4–6	Flashing light	Flashing light (B132)	TS-1 of the touch panel	V1
J4 7–10	PLC DC common	PLC-5 DCM module (B132-13)	PLC DC COM	V1
J5 1–3	HVPS current monitor	X513 (to the panel and the PLC-5)	TS-3 power-supply monitors (SPR02923; SPR02924)	V1
J5 4–6	HVPS voltage monitor	X513 (to the panel and the PLC-5)	TS-3 power-supply monitors (SPR02923; SPR02924)	V1
J5 7–9	Oil level LEV-3 *The drawing points the SPEAR 2 branch to sheet 2 (E17)*	SPR02946 from the oil-level switch	TS-6 terminals 7 and 8	V1
J6 1–2	Switch-tank flow switch *Terminal 3 marked N/C*	Flow switch beside the cable switch tank (B514)	TS-4 terminals 13 and 14	V1
BNC-1	Klystron arc	SPR02947 from the termination tank	BNC-1 of the left-side interface board	V1
Fibers	Three panel fibers *See the fiber table; re-patched to the unit in service at a switch-over*	RF Local Display Panel U11; U19; U20	Right-side board J2 and left-side board J2 and J3	V1

6.5 RF MPS PLC-5 Inputs and Outputs

The RF MPS PLC-5 reads and drives the station through 20 I/O modules in two 1771 chassis, X430 and X420, in rack B132-13 (Table 14): fourteen thermocouple modules and two each of analog input, digital input and digital output modules, each with its own wiring diagram, WD-340-330-07 to -26. Tables 15 to 18 list all 272 points: the module terminal, the signal, the far end and the cross-connects on the way (*Through*), and the name the software uses. That name is the EPICS record of a thermocouple or analog channel, read through the 1771-DCM, and the symbol of a digital point in the decoded PLC-5 program (*Symbol*). A point is at level V1 when the drawing and the software agree, or when an erratum explains the difference; consecutive spare points are listed as one row.

Two numbering rules matter. An analog channel is numbered in the order the module and EPICS read it, so its terminal differs: terminals 5, 10 and 15 are returns, and channel 9 is on terminal 11. And the drawings number some cross-connects differently: X519 (E7), and X530 and the panel contacts, for which the tables follow WD-340-330-02 (E6). Reading the module drawings against the program found three more errata: two thermocouple modules labeled with the wrong slot (E20), five input labels one row late (E21) and an input drawn spare that the program reads (E22).

At I4 the conversion kit puts these field-wiring arms on the ControlLogix (Doc M), so each row maps a terminal to its tag; FC24 gathers the points still open. At I2 the Interface Chassis takes over the panel lines marked in the Chassis column (P1 to P12, N3 and N4) and drives the RF power level on analog input 2, channel 10 (N6, Doc C).

Table 14: I/O modules of the RF MPS PLC-5.

Module	Catalog	Location and address	Drawing	Content and notes
TC1	1771-IXE	X430 module 1 BT rack 0 group 0	WD-340-330-07	Klystron water, body, collector and window air; HCW supply *Far ends XWTR-TB2 1 to 8 in the klystron water rack, on 8 pairs of a 12-pair cable; EPICS DCM table 2 words 4 to 11*
TC2	1771-IXE	X430 module 2 BT rack 0 group 1	WD-340-330-08	Station: VXI crate supply, ambient air, RF drive amplifier; HCW returns of the circulator and waveguide loads *Channels 5 to 8 on XWTR-TB2 9 to 12; EPICS DCM table 2 words 12 to 19*
TC3	1771-IXE	X430 module 3 BT rack 0 group 2	WD-340-330-09	Cavity A body 1 to 8 *Far ends CAVAJB-TB2 1 to 8; EPICS DCM table 2 words 20 to 27*
TC4	1771-IXE	X430 module 4 BT rack 0 group 3	WD-340-330-10	Cavity A body 9 to 14, fixed and movable tuner *Far ends CAVAJB-TB2 9 to 16; EPICS DCM table 2 words 28 to 35*
TC5	1771-IXE	X430 module 5 BT rack 0 group 4	WD-340-330-11	Cavity A tuner bellows, HOM loads, LCW return and supply, waveguide HOM load *Far ends CAVAJB-TB2 17 to 24; EPICS DCM table 2 words 36 to 43*
TC6	1771-IXE	X430 module 6 BT rack 0 group 5	WD-340-330-12	Cavity B body 1 to 8 *Far ends CAVBJB-TB2 1 to 8; EPICS DCM table 2 words 44 to 51*

Module	Catalog	Location and address	Drawing	Content and notes
TC7	1771-IXE	X430 module 7 BT rack 0 group 6	WD-340-330-13	Cavity B body 9 to 14, fixed and movable tuner *Far ends CAVBJB-TB2 9 to 16; EPICS DCM table 2 words 52 to 59*
TC8	1771-IXE	X430 module 8 BT rack 0 group 7	WD-340-330-14	Cavity B tuner bellows, HOM loads, LCW return and supply, waveguide HOM load *Far ends CAVBJB-TB2 17 to 24; EPICS DCM table 3 words 4 to 11*
TC9	1771-IXE	X420 module 1 BT rack 2 group 0	WD-340-330-15	Cavity C body 1 to 8 *Far ends CAVCJB-TB2 1 to 8; EPICS DCM table 3 words 12 to 19*
TC10	1771-IXE	X420 module 2 BT rack 2 group 1	WD-340-330-16	Cavity C body 9 to 14, fixed and movable tuner *Far ends CAVCJB-TB2 9 to 16; EPICS DCM table 3 words 20 to 27*
TC11	1771-IXE	X420 module 3 (drawn MODULE 11, E20) BT rack 2 group 2	WD-340-330-17	Cavity C tuner bellows, HOM loads, LCW return and supply, waveguide HOM load *Far ends CAVCJB-TB2 17 to 24; EPICS DCM table 3 words 28 to 35*
TC12	1771-IXE	X420 module 4 BT rack 2 group 3	WD-340-330-18	Cavity D body 1 to 8 *Far ends CAVDJB-TB2 1 to 8; EPICS DCM table 3 words 36 to 43*
TC13	1771-IXE	X420 module 5 (drawn MODULE 25, E20) BT rack 2 group 4	WD-340-330-19	Cavity D body 9 to 14, fixed and movable tuner *Far ends CAVDJB-TB2 9 to 16; EPICS DCM table 3 words 44 to 51*
TC14	1771-IXE	X420 module 6 BT rack 2 group 5	WD-340-330-20	Cavity D tuner bellows, HOM loads, LCW return and supply, waveguide HOM load *Far ends CAVDJB-TB2 17 to 24; EPICS DCM table 3 words 52 to 59*
AI1	Analog input, 16 single-ended channels (catalog not drawn)	X430 slot 9 (drawn RACK 4 SLOT 9) BT rack 1 group 0	WD-340-330-21	Cavities A and B; filament, main focus and bucking supplies; klystron vacuum; circulator *Through X516; terminals 5, 10 and 15 return, 20 and 21 common; channel n in EPICS DCM word n + 3*
AI2	Analog input, 16 single-ended channels (catalog not drawn)	X430 slot 10 (drawn RACK 4 SLOT 10) BT rack 1 group 1	WD-340-330-22	Cavities C and D; tuner 24 V bulk supply; RF power level; HVPS voltage and current *Through X517; channels 13 to 16 spare, tied to the return bus; channel n in EPICS DCM word n + 19*
DI1	1771-IBN	X420 slot 9 (drawn RACK X4 MODULE 9) I:030, I:031	WD-340-330-23	Klystron, circulator and waveguide interlocks; requests from the panel *Through X519; terminals 9, 10, 19, 20, 29, 30, 39 and 40 are the -DC return. X519 numbers as drawn on WD-340-330-23, except on the panel lines, which follow WD-340-330-02; WD-340-330-02 and -33 number X519 by module terminal (E7)*

Module	Catalog	Location and address	Drawing	Content and notes
DO1	1771-OBN	X420 slot 10 (drawn RACK X4 MODULE 10) O:030, O:031	WD-340-330-25	Interlock summaries and commands to the panel; filament, focus and blower commands *Through X522; +DC on terminals 1, 11, 21 and 31 from X518; commons on 10, 20, 30 and 40*
DI2	1771-IBN	X420 slot 11 (drawn RACK X4 MODULE 11) I:032, I:033	WD-340-330-24	Cavity flow, vacuum and ion-pump interlocks *Through X520; terminals 9, 10, 19, 20, 29, 30, 39 and 40 are the -DC return*
DO2	1771-OBN	X420 slot 12 (drawn RACK X4 MODULE 12) O:032, O:033	WD-340-330-26	All outputs drawn spare *Through X523 (terminal n on X523 n); +DC on 1, 11, 21 and 31; commons on 10, 20, 30 and 40*

Table 15: Thermocouple inputs of the RF MPS PLC-5 (1771-IXE; terminals +/−).

Ch.	Terminals	Signal	Far end	EPICS record	V
TC1: X430 module 1; BT rack 0 group 0; WD-340-330-07					
1	18/17	KLYSTRON WATER SUPPLY	XWTR-TB2 1 (thermowell)	SRF1:KLYSWATR:TEMP	V1
2	16/15	KLYSTRON BODY 1 *EPICS: body 1 water delta*	XWTR-TB2 2 (thermowell)	SRF1:KLYSBD01WATR:TEMP	V1
3	14/13	KLYSTRON BODY 2 *EPICS: body 2 water delta*	XWTR-TB2 3 (thermowell)	SRF1:KLYSBD02WATR:TEMP	V1
4	12/11	KLYSTRON BODY 3 *EPICS: body 3 water delta*	XWTR-TB2 4 (thermowell)	SRF1:KLYSBD03WATR:TEMP	V1
5	8/7	KLYSTRON COLLECTOR *EPICS: collector water delta*	XWTR-TB2 5 (thermowell)	SRF1:KLYSCOLLWATR:TEMP	V1
6	6/5	KLYSTRON WINDOW AIR IN *On the window air supply*	XWTR-TB2 6	SRF1:KLYSWNDWAIR:TEMP	V1
7	4/3	KLYSTRON WINDOW AIR OUT *On the window air return*	XWTR-TB2 7	SRF1:KLYSWNDWAIRI:TEMP	V1
8	2/1	HCW SUPPLY	XWTR-TB2 8 (thermowell)	SRF1:STNHCW:TEMP	V1
TC2: X430 module 2; BT rack 0 group 1; WD-340-330-08					
1	18/17	VXI POWER SUPPLY	Thermocouple in blue rack 7	SRF1:STNCRATEPS:TEMP	V1
2	16/15	AMBIENT AIR	Thermocouple above rack 4	SRF1:STNAMBAIR:TEMP	V1
3	14/13	HVPS oil temperature; drawn SPARE *E14*	Not drawn	SRF1:HVPSOIL:TEMP	V1
4	12/11	RF AMPLIFIER	Thermocouple in blue rack 7	SRF1:KLYSDRIVAMPL:TEMP	V1

Ch.	Terminals	Signal	Far end	EPICS record	V
5	8/7	CIRCULATOR LOAD HCW RETURN	XWTR-TB2 9 (thermowell)	SRF1:CIRCHCW:TEMP	V1
6	6/5	W/G LOAD #1 HCW RETURN	XWTR-TB2 10 (thermowell)	SRF1:WG01HCW:TEMP	V1
7	4/3	W/G LOAD #2 HCW RETURN	XWTR-TB2 11 (thermowell)	SRF1:WG02HCW:TEMP	V1
8	2/1	W/G LOAD #3 HCW RETURN	XWTR-TB2 12 (thermowell)	SRF1:WG03HCW:TEMP	V1
TC3: X430 module 3; BT rack 0 group 2; WD-340-330-09					
1	18/17	CAVITY A BODY 1	CAVAJB-TB2 1	SRF1:CAV1BD01:TEMP	V1
2	16/15	CAVITY A BODY 2	CAVAJB-TB2 2	SRF1:CAV1BD02:TEMP	V1
3	14/13	CAVITY A BODY 3	CAVAJB-TB2 3	SRF1:CAV1BD03:TEMP	V1
4	12/11	CAVITY A BODY 4	CAVAJB-TB2 4	SRF1:CAV1BD04:TEMP	V1
5	8/7	CAVITY A BODY 5	CAVAJB-TB2 5	SRF1:CAV1BD05:TEMP	V1
6	6/5	CAVITY A BODY 6	CAVAJB-TB2 6	SRF1:CAV1BD06:TEMP	V1
7	4/3	CAVITY A BODY 7	CAVAJB-TB2 7	SRF1:CAV1BD07:TEMP	V1
8	2/1	CAVITY A BODY 8	CAVAJB-TB2 8	SRF1:CAV1BD08:TEMP	V1
TC4: X430 module 4; BT rack 0 group 3; WD-340-330-10					
1	18/17	CAVITY A BODY 9	CAVAJB-TB2 9	SRF1:CAV1BD09:TEMP	V1
2	16/15	CAVITY A BODY 10	CAVAJB-TB2 10	SRF1:CAV1BD10:TEMP	V1
3	14/13	CAVITY A BODY 11	CAVAJB-TB2 11	SRF1:CAV1BD11:TEMP	V1
4	12/11	CAVITY A BODY 12	CAVAJB-TB2 12	SRF1:CAV1BD12:TEMP	V1
5	8/7	CAVITY A BODY 13	CAVAJB-TB2 13	SRF1:CAV1BD13:TEMP	V1
6	6/5	CAVITY A BODY 14	CAVAJB-TB2 14	SRF1:CAV1BD14:TEMP	V1
7	4/3	CAVITY A FIXED TUNER	CAVAJB-TB2 15	SRF1:CAV1TUNRFX:TEMP	V1
8	2/1	CAVITY A MOVABLE TUNER	CAVAJB-TB2 16	SRF1:CAV1TUNRMV:TEMP	V1
TC5: X430 module 5; BT rack 0 group 4; WD-340-330-11					
1	18/17	CAVITY A TUNER BELLOWS	CAVAJB-TB2 17	SRF1:CAV1TUNRBL:TEMP	V1
2	16/15	CAVITY A HOM LOAD #1	CAVAJB-TB2 18	SRF1:CAV1HOM1:TEMP	V1
3	14/13	CAVITY A HOM LOAD #2	CAVAJB-TB2 19	SRF1:CAV1HOM2:TEMP	V1
4	12/11	CAVITY A HOM LOAD #3	CAVAJB-TB2 20	SRF1:CAV1HOM3:TEMP	V1
5	8/7	CAVITY A LCW RETURN	CAVAJB-TB2 21	SRF1:CAV1WATRRTN:TEMP	V1
6	6/5	CAVITY A LCW SUPPLY	CAVAJB-TB2 22	SRF1:CAV1WATR:TEMP	V1
7	4/3	CAVITY A W/G HOM LOAD	CAVAJB-TB2 23	SRF1:CAV1WGFERR:TEMP	V1
8	2/1	Ion-pump temperature; drawn SPARE *E15: read live by EPICS*	CAVAJB-TB2 24	SRF1:CAV1IONP:TEMP	V1
TC6: X430 module 6; BT rack 0 group 5; WD-340-330-12					
1	18/17	CAVITY B BODY 1	CAVBJB-TB2 1	SRF1:CAV2BD01:TEMP	V1
2	16/15	CAVITY B BODY 2	CAVBJB-TB2 2	SRF1:CAV2BD02:TEMP	V1
3	14/13	CAVITY B BODY 3	CAVBJB-TB2 3	SRF1:CAV2BD03:TEMP	V1

Ch.	Terminals	Signal	Far end	EPICS record	V
4	12/11	CAVITY B BODY 4	CAVBJB-TB2 4	SRF1:CAV2BD04:TEMP	V1
5	8/7	CAVITY B BODY 5	CAVBJB-TB2 5	SRF1:CAV2BD05:TEMP	V1
6	6/5	CAVITY B BODY 6	CAVBJB-TB2 6	SRF1:CAV2BD06:TEMP	V1
7	4/3	CAVITY B BODY 7	CAVBJB-TB2 7	SRF1:CAV2BD07:TEMP	V1
8	2/1	CAVITY B BODY 8	CAVBJB-TB2 8	SRF1:CAV2BD08:TEMP	V1
TC7: X430 module 7; BT rack 0 group 6; WD-340-330-13					
1	18/17	CAVITY B BODY 9	CAVBJB-TB2 9	SRF1:CAV2BD09:TEMP	V1
2	16/15	CAVITY B BODY 10	CAVBJB-TB2 10	SRF1:CAV2BD10:TEMP	V1
3	14/13	CAVITY B BODY 11	CAVBJB-TB2 11	SRF1:CAV2BD11:TEMP	V1
4	12/11	CAVITY B BODY 12	CAVBJB-TB2 12	SRF1:CAV2BD12:TEMP	V1
5	8/7	CAVITY B BODY 13	CAVBJB-TB2 13	SRF1:CAV2BD13:TEMP	V1
6	6/5	CAVITY B BODY 14	CAVBJB-TB2 14	SRF1:CAV2BD14:TEMP	V1
7	4/3	CAVITY B FIXED TUNER	CAVBJB-TB2 15	SRF1:CAV2TUNRFX:TEMP	V1
8	2/1	CAVITY B MOVABLE TUNER	CAVBJB-TB2 16	SRF1:CAV2TUNRMV:TEMP	V1
TC8: X430 module 8; BT rack 0 group 7; WD-340-330-14					
1	18/17	CAVITY B TUNER BELLOWS	CAVBJB-TB2 17	SRF1:CAV2TUNRBL:TEMP	V1
2	16/15	CAVITY B HOM LOAD #1	CAVBJB-TB2 18	SRF1:CAV2HOM1:TEMP	V1
3	14/13	CAVITY B HOM LOAD #2	CAVBJB-TB2 19	SRF1:CAV2HOM2:TEMP	V1
4	12/11	CAVITY B HOM LOAD #3	CAVBJB-TB2 20	SRF1:CAV2HOM3:TEMP	V1
5	8/7	CAVITY B LCW RETURN	CAVBJB-TB2 21	SRF1:CAV2WATRRTN:TEMP	V1
6	6/5	CAVITY B LCW SUPPLY	CAVBJB-TB2 22	SRF1:CAV2WATR:TEMP	V1
7	4/3	CAVITY B W/G HOM LOAD	CAVBJB-TB2 23	SRF1:CAV2WGFERR:TEMP	V1
8	2/1	Ion-pump temperature; drawn SPARE *E15: read live by EPICS*	CAVBJB-TB2 24	SRF1:CAV2IONP:TEMP	V1
TC9: X420 module 1; BT rack 2 group 0; WD-340-330-15					
1	18/17	CAVITY C BODY 1	CAVCJB-TB2 1	SRF1:CAV3BD01:TEMP	V1
2	16/15	CAVITY C BODY 2	CAVCJB-TB2 2	SRF1:CAV3BD02:TEMP	V1
3	14/13	CAVITY C BODY 3	CAVCJB-TB2 3	SRF1:CAV3BD03:TEMP	V1
4	12/11	CAVITY C BODY 4	CAVCJB-TB2 4	SRF1:CAV3BD04:TEMP	V1
5	8/7	CAVITY C BODY 5	CAVCJB-TB2 5	SRF1:CAV3BD05:TEMP	V1
6	6/5	CAVITY C BODY 6	CAVCJB-TB2 6	SRF1:CAV3BD06:TEMP	V1
7	4/3	CAVITY C BODY 7	CAVCJB-TB2 7	SRF1:CAV3BD07:TEMP	V1
8	2/1	CAVITY C BODY 8	CAVCJB-TB2 8	SRF1:CAV3BD08:TEMP	V1
TC10: X420 module 2; BT rack 2 group 1; WD-340-330-16					
1	18/17	CAVITY C BODY 9	CAVCJB-TB2 9	SRF1:CAV3BD09:TEMP	V1
2	16/15	CAVITY C BODY 10	CAVCJB-TB2 10	SRF1:CAV3BD10:TEMP	V1
3	14/13	CAVITY C BODY 11	CAVCJB-TB2 11	SRF1:CAV3BD11:TEMP	V1
4	12/11	CAVITY C BODY 12	CAVCJB-TB2 12	SRF1:CAV3BD12:TEMP	V1

Ch.	Terminals	Signal	Far end	EPICS record	V
5	8/7	CAVITY C BODY 13	CAVCJB-TB2 13	SRF1:CAV3BD13:TEMP	V1
6	6/5	CAVITY C BODY 14	CAVCJB-TB2 14	SRF1:CAV3BD14:TEMP	V1
7	4/3	CAVITY C FIXED TUNER	CAVCJB-TB2 15	SRF1:CAV3TUNRFX:TEMP	V1
8	2/1	CAVITY C MOVABLE TUNER	CAVCJB-TB2 16	SRF1:CAV3TUNRMV:TEMP	V1
TC11: X420 module 3 (drawn MODULE 11, E20); BT rack 2 group 2; WD-340-330-17					
1	18/17	CAVITY C TUNER BELLOWS	CAVCJB-TB2 17	SRF1:CAV3TUNRBL:TEMP	V1
2	16/15	CAVITY C HOM LOAD #1	CAVCJB-TB2 18	SRF1:CAV3HOM1:TEMP	V1
3	14/13	CAVITY C HOM LOAD #2	CAVCJB-TB2 19	SRF1:CAV3HOM2:TEMP	V1
4	12/11	CAVITY C HOM LOAD #3	CAVCJB-TB2 20	SRF1:CAV3HOM3:TEMP	V1
5	8/7	CAVITY C LCW RETURN	CAVCJB-TB2 21	SRF1:CAV3WATRRTN:TEMP	V1
6	6/5	CAVITY C LCW SUPPLY	CAVCJB-TB2 22	SRF1:CAV3WATR:TEMP	V1
7	4/3	CAVITY C W/G HOM LOAD	CAVCJB-TB2 23	SRF1:CAV3WGFERR:TEMP	V1
8	2/1	Ion-pump temperature; drawn SPARE *E15: read live by EPICS*	CAVCJB-TB2 24	SRF1:CAV3IONP:TEMP	V1
TC12: X420 module 4; BT rack 2 group 3; WD-340-330-18					
1	18/17	CAVITY D BODY 1	CAVDJB-TB2 1	SRF1:CAV4BD01:TEMP	V1
2	16/15	CAVITY D BODY 2	CAVDJB-TB2 2	SRF1:CAV4BD02:TEMP	V1
3	14/13	CAVITY D BODY 3	CAVDJB-TB2 3	SRF1:CAV4BD03:TEMP	V1
4	12/11	CAVITY D BODY 4	CAVDJB-TB2 4	SRF1:CAV4BD04:TEMP	V1
5	8/7	CAVITY D BODY 5	CAVDJB-TB2 5	SRF1:CAV4BD05:TEMP	V1
6	6/5	CAVITY D BODY 6	CAVDJB-TB2 6	SRF1:CAV4BD06:TEMP	V1
7	4/3	CAVITY D BODY 7	CAVDJB-TB2 7	SRF1:CAV4BD07:TEMP	V1
8	2/1	CAVITY D BODY 8	CAVDJB-TB2 8	SRF1:CAV4BD08:TEMP	V1
TC13: X420 module 5 (drawn MODULE 25, E20); BT rack 2 group 4; WD-340-330-19					
1	18/17	CAVITY D BODY 9	CAVDJB-TB2 9	SRF1:CAV4BD09:TEMP	V1
2	16/15	CAVITY D BODY 10	CAVDJB-TB2 10	SRF1:CAV4BD10:TEMP	V1
3	14/13	CAVITY D BODY 11	CAVDJB-TB2 11	SRF1:CAV4BD11:TEMP	V1
4	12/11	CAVITY D BODY 12	CAVDJB-TB2 12	SRF1:CAV4BD12:TEMP	V1
5	8/7	CAVITY D BODY 13	CAVDJB-TB2 13	SRF1:CAV4BD13:TEMP	V1
6	6/5	CAVITY D BODY 14	CAVDJB-TB2 14	SRF1:CAV4BD14:TEMP	V1
7	4/3	CAVITY D FIXED TUNER	CAVDJB-TB2 15	SRF1:CAV4TUNRFX:TEMP	V1
8	2/1	CAVITY D MOVABLE TUNER	CAVDJB-TB2 16	SRF1:CAV4TUNRMV:TEMP	V1
TC14: X420 module 6; BT rack 2 group 5; WD-340-330-20					
1	18/17	CAVITY D TUNER BELLOWS	CAVDJB-TB2 17	SRF1:CAV4TUNRBL:TEMP	V1
2	16/15	CAVITY D HOM LOAD #1	CAVDJB-TB2 18	SRF1:CAV4HOM1:TEMP	V1
3	14/13	CAVITY D HOM LOAD #2	CAVDJB-TB2 19	SRF1:CAV4HOM2:TEMP	V1
4	12/11	CAVITY D HOM LOAD #3	CAVDJB-TB2 20	SRF1:CAV4HOM3:TEMP	V1
5	8/7	CAVITY D LCW RETURN	CAVDJB-TB2 21	SRF1:CAV4WATRRTN:TEMP	V1

Ch.	Terminals	Signal	Far end	EPICS record	V
6	6/5	CAVITY D LCW SUPPLY	CAVDJB-TB2 22	SRF1:CAV4WATR:TEMP	V1
7	4/3	CAVITY D W/G HOM LOAD	CAVDJB-TB2 23	SRF1:CAV4WGFERR:TEMP	V1
8	2/1	Ion-pump temperature; drawn SPARE *E15: read live by EPICS*	CAVDJB-TB2 24	SRF1:CAV4IONP:TEMP	V1

Table 16: Analog inputs of the RF MPS PLC-5.

Ch.	Term.	Signal	Far end	Through	EPICS record	Chassis	V
AI1: X430 slot 9 (drawn RACK 4 SLOT 9); BT rack 1 group 0; WD-340-330-21							
1	1	CAVA TUNER POS	CAVAJB-TB01 18	X507 18; X516 1	SRF1:CAV1TUNR: POSN	–	V1
2	2	CAVA VACUUM	X328-AO +	X528 19; X516 2	SRF1:CAV1:VACM	–	V1
3	3	CAVA PUMP I	X319-S1-J3 05	X528 21; X516 3	SRF1:CAV1IONP: CURR	–	V1
4	4	CAVA I/R	CAVAJB-TB01 21	X507 21; X516 4	SRF1:CAV1WNDW: TEMP	–	V1
5	6	CAVB TUNER POS	CAVBJB-TB01 18	X508 18; X516 6	SRF1:CAV2TUNR: POSN	–	V1
6	7	CAVB VACUUM	X326-AO +	X528 44; X516 7	SRF1:CAV2:VACM	–	V1
7	8	CAVB PUMP I	X319-S2-J3 05	X528 46; X516 8	SRF1:CAV2IONP: CURR	–	V1
8	9	CAVB I/R	CAVBJB-TB01 21	X508 21; X516 9	SRF1:CAV2WNDW: TEMP	–	V1
9	11	FILAMENT V *WD-340-330-28 agrees*	X610-J1 E (filament chassis 340-311)	X525 25; X516 11	SRF1:FILAMENT:VOLT	–	V1
10	12	FILAMENT I *WD-340-330-28 agrees*	X610-J1 G (filament chassis 340-311)	X525 27; X516 12	SRF1:FILAMENT: CURR	–	V1
11	13	FOCUS V	X616-J1 19 (main focus supply)	X525 05; X516 13	SRF1:SOLNMAIN: VOLT	–	V1
12	14	FOCUS I	X616-J1 20 (main focus supply)	X525 07; X516 14	SRF1:SOLNMAIN: CURR	–	V1
13	16	BUCKING V	X621-J1 A (bucking supply)	X525 15; X516 16	SRF1:SOLNBUCK: VOLT	–	V1
14	17	BUCKING I	X621-J1 C (bucking supply)	X525 17; X516 17	SRF1:SOLNBUCK: CURR	–	V1

Ch.	Term.	Signal	Far end	Through	EPICS record	Chassis	V
15	18	KLYSTRON VACUUM *The far end is drawn NOT USED; EPICS still has the record (FC24)*	Not used	X525 31; X516 18	SRF1:KLYS:VACM	–	V1
16	19	CIRCULATOR I	X625-CC 01 (circulator controller)	X526 01; X516 19	SRF1:CIRC:CURR	–	V1
AI2: X430 slot 10 (drawn RACK 4 SLOT 10); BT rack 1 group 1; WD-340-330-22							
1	1	CAVC TUNER POS	CAVCJB-TB01 18	X509 18; X517 1	SRF1:CAV3TUNR: POSN	–	V1
2	2	CAVC VACUUM	X324-AO +	X529 19; X517 2	SRF1:CAV3:VACM	–	V1
3	3	CAVC PUMP I	X319-S3-J3 05	X529 21; X517 3	SRF1:CAV3IONP: CURR	–	V1
4	4	CAVC I/R	CAVCJB-TB01 21	X509 21; X517 4	SRF1:CAV3WNDW: TEMP	–	V1
5	6	CAVD TUNER POS	CAVDJB-TB01 18	X510 18; X517 6	SRF1:CAV4TUNR: POSN	–	V1
6	7	CAVD VACUUM	X322-AO +	X529 44; X517 7	SRF1:CAV4:VACM	–	V1
7	8	CAVD PUMP I	X319-S4-J3 05	X529 46; X517 8	SRF1:CAV4IONP: CURR	–	V1
8	9	CAVD I/R	CAVDJB-TB01 21	X510 21; X517 9	SRF1:CAV4WNDW: TEMP	–	V1
9	11	24V BULK PS	X623-J1 A (tuner 24 V bulk supply)	X525 41; X517 11	SRF1:CAVTUNR:VOLT	–	V1
10	12	RF PWR LEVEL *Level and detector transfer curve: FC4*	X7-ARC-RFAMP A (detector output of the Fast Interlock Chassis)	X513 01; X517 12	SRF1:KLYSPLC: POWER	N6	V1
11	13	HVPS V *WD-340-330-02 agrees*	Panel J2 3 (HVPS Vout)	X530 3; X517 13	SRF1:HVPSPLC:VOLT	N3	V1
12	14	HVPS I *WD-340-330-02 agrees*	Panel J2 4 (HVPS Iout)	X530 4; X517 14	SRF1:HVPSPLC:CURR	N4	V1
13	16	SPARE *No EPICS record*	Tied to the return bus	X517 16	–	–	V1
14	17	SPARE *No EPICS record*	Tied to the return bus	X517 17	–	–	V1
15	18	SPARE *No EPICS record*	Tied to the return bus	X517 18	–	–	V1
16	19	SPARE *No EPICS record*	Tied to the return bus	X517 19	–	–	V1

Table 17: Digital inputs of the RF MPS PLC-5 (1771-IBN).

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
DI1: X420 slot 9 (drawn RACK X4 MODULE 9); I:030, I:031; WD-340-330-23							
I:030/00	1	RESET *WD-340-330-23 draws X330-J1 11, X530 11, X519 61 (E6, E7)*	Panel J2 23	X530 23; X519 1	LOCAL_ RESET	P10	V1
I:030/01	2	LOCAL/REMOTE *WD-340-330-23 draws X519 62 (E7)*	Panel J2 12	X530 12; X519 2	LOCAL_ REMOTE	P12 (open)	V1
I:030/02	3	MPS *Waveguide-air MPS input*	WG-AIR 2	X515 3; X519 63	MPS	–	V1
I:030/03	4	KLYSTRON COLLECTOR FLOW *Flow switch: WD-340-330-31*	WTR-TB1 8	X505 8; X519 64	COLL_FLOW	–	V1
I:030/04	5	KLYSTRON BODY 1 FLOW *Flow switch: WD-340-330-31*	WTR-TB1 9	X505 9; X519 65	BODY_1_ FLOW	–	V1
I:030/05	6	KLYSTRON BODY 2 FLOW *Flow switch: WD-340-330-31*	WTR-TB1 10	X505 10; X519 66	BODY_2_ FLOW	–	V1
I:030/06	7	KLYSTRON BODY 3 FLOW *X519 76 is out of sequence as drawn*	WTR-TB1 11	X505 11; X519 76	BODY_3_ FLOW	–	V1
I:030/07	8	HCW 1 WATER FLOW	WTR-TB1 12	X505 12; X519 67	HCW_1_ FLOW	–	V1
I:030/10	11	HCW 2 FLOW	WTR-TB1 13	X505 13; X519 68	HCW_2_ FLOW	–	V1
I:030/11	12	KLYSTRON VACUUM	KLY-VAC J2 2	X525 6; X519 69	KLY_VAC_I	–	V1
I:030/12	13	Klystron window air flow *Drawn CIRCULATOR WATER INPUT TEMP (E21); WD-340-330-33 agrees with the program and draws XBLWR-MTR H, X512 07, X519 13 (E7)*	BLWR 2 (window air flow meter)	X512 2; X519 70	KLY_AIR_ FLOW	–	V1
I:030/13	14	Circulator water input temperature *Drawn CIRCULATOR WATER DELTA TEMP (E21); WD-340-330-34 agrees with the program*	X625-CC 3	X526 3; X519 71	CIRC_ WATER_IN	–	V1

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
I:030/14	15	Circulator water delta temperature *Drawn WAVEGUIDE AIR PRESSURE (E21); WD-340-330-34 agrees with the program*	X625-CC 4	X526 4; X519 72	CIRC_DELTA	–	V1
I:030/15	16	Waveguide air pressure *Drawn CIRCULATOR FLOW (E21)*	WG-AIR-SW 2	X515 4; X519 73	WG_AIR	–	V1
I:030/16	17	Circulator flow *Drawn LER/HER STATION TYPE (E21); WD-340-330-31 agrees with the program*	WTR-TB1 14	X505 14; X519 74	CIRC_FLOW	–	V1
I:030/17	18	SPARE	Not connected	X519 75	CIRC_FERR_ TEMP, not used	–	V1
I:031/00	21	SPARE	Not connected	X519 1	VXI_RESET, not used	–	V1
I:031/01	22	FILAMENT ON/OFF *WD-340-330-23 draws X330-J1 1, X530 1, X519 2 (E6, E7)*	Panel J2 21	X530 21; X519 22	FIL_ON_OFF	P8	V1
I:031/02	23	FILAMENT TIMER BYPASS *WD-340-330-23 draws X330-J1 2, X530 2, X519 3 (E6, E7)*	Panel J2 22	X530 22; X519 23	FIL_BYP_DI	P9	V1
I:031/03	24	KLYSTRON FOCUSING ON/OFF *The panel calls it the online/offline request; WD-340-330-23 draws X330-J1 3, X530 3, X519 4 (E6, E7)*	Panel J2 18	X530 18; X519 24	KLY_FOC_DI	P7	V1
I:031/04	25	VXI summary input, held on *Drawn SPARE (E22); the program latches NIRP_IN if it drops*	+24 V (X518 1)	X519 5	VXI_SUM_DI	–	V1
I:031/05	26	MAIN FOCUS INTERNAL (N/C)	X616-TB1 1	X525 1; X519 6	FOCUS_1_ INT_DI	–	V1
I:031/06	27	BUCKING INTERNAL (N/C)	X621-TB1 1	X525 2; X519 7	FOCUS_2_ INT_DI, not used	–	V1
I:031/07	28	ONLINE/OFFLINE MODE *WD-340-330-23 draws X330-J1 4, X530 4, X519 8 (E6, E7)*	Panel J2 25	X530 25; X519 28	ONLINE	P11 (held on)	V1

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
I:031/10	31	SPARE	Not connected	X519 10	Unnamed; read by rung DIGI_IN1:002	–	V1
I:031/11–17	32–38	SPARE	Not connected	X519 14–20	–	–	V1
DI2: X420 slot 11 (drawn RACK X4 MODULE 11); I:032, I:033; WD-340-330-24							
I:032/00	1	CAVA FLOW	CAVAJB-TB01 08	X507 08; X520 1	CAVA_FLOW_I	–	V1
I:032/01	2	CAVA VACUUM	X328-PC T	X528 18; X520 2	CAVA_VAC_I	–	V1
I:032/02	3	SPARE	Not connected	X520 3	CAVA_FIL_I, rungs disabled	–	V1
I:032/03	4	SPARE	Not connected	X520 4	CAVA_DEGASS_I, rungs disabled	–	V1
I:032/04	5	CAVA PUMP I	X319-S1-J3 20	X528 24; X520 5	CAVA_PMP_I	–	V1
I:032/05	6	SPARE	Not connected	X520 6	CAVA_GAUGE_I, rungs disabled	–	V1
I:032/06–07	7–8	SPARE	Not connected	X520 7–8	–	–	V1
I:032/10	11	CAVB FLOW	CAVBJB-TB01 08	X508 08; X520 11	CAVB_FLOW_I	–	V1
I:032/11	12	CAVB VACUUM	X326-PC T	X528 43; X520 12	CAVB_VAC_I	–	V1
I:032/12	13	SPARE	Not connected	X520 13	CAVB_FIL_I, rungs disabled	–	V1
I:032/13	14	SPARE	Not connected	X520 14	CAVB_DEGASS_I, rungs disabled	–	V1
I:032/14	15	CAVB PUMP I *Pin 1 as drawn; cavities A, C and D use pin 20 (FC24)*	X319-S2-J3 1	X528 49; X520 15	CAVB_PMP_I	–	V1
I:032/15	16	SPARE	Not connected	X520 16	CAVB_GAUGE_I, rungs disabled	–	V1
I:032/16–17	17–18	SPARE	Not connected	X520 17–18	–	–	V1
I:033/00	21	CAVC FLOW	CAVCJB-TB01 08	X509 08; X520 21	CAVC_FLOW_I	–	V1
I:033/01	22	CAVC VACUUM	X324-PC T	X529 18; X520 22	CAVC_VAC_I	–	V1
I:033/02	23	SPARE	Not connected	X520 23	CAVC_FIL_I, rungs disabled	–	V1
I:033/03	24	SPARE	Not connected	X520 24	CAVC_DEGASS_I, rungs disabled	–	V1

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
I:033/04	25	CAVC PUMP I	X319-S3-J3 20	X529 24; X520 25	CAVC_PMP_I	–	V1
I:033/05	26	SPARE	Not connected	X520 26	CAVC_ GAUGE_I, rungs disabled	–	V1
I:033/06–07	27–28	SPARE	Not connected	X520 27–28	–	–	V1
I:033/10	31	CAVD FLOW	CAVDJB-TB01 08	X510 08; X520 31	CAVD_FLOW_ I	–	V1
I:033/11	32	CAVD VACUUM	X322-PC T	X529 43; X520 32	CAVD_VAC_I	–	V1
I:033/12	33	SPARE	Not connected	X520 33	CAVD_FIL_I, rungs disabled	–	V1
I:033/13	34	SPARE	Not connected	X520 34	CAVD_ DEGASS_I, rungs disabled	–	V1
I:033/14	35	CAVD PUMP I	X319-S4-J3 20	X529 49; X520 35	CAVD_PMP_I	–	V1
I:033/15	36	SPARE	Not connected	X520 36	CAVD_ GAUGE_I, rungs disabled	–	V1
I:033/16–17	37–38	SPARE	Not connected	X520 37–38	–	–	V1

Table 18: Digital outputs of the RF MPS PLC-5 (1771-OBN).

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
DO1: X420 slot 10 (drawn RACK X4 MODULE 10); O:030, O:031; WD-340-330-25							
O:030/00	2	CAVA INTERLOCK SUM	Panel J3, WD-02 contact 13 (E4)	X530 38; X522 2	CAVA_SUM_ DO	display only	V1
O:030/01	3	CAVB INTERLOCK SUM	Panel J3, WD-02 contact 25 (E4)	X530 50; X522 3	CAVB_SUM_ DO	display only	V1
O:030/02	4	CAVC INTERLOCK SUM	Panel J3, WD-02 contact 12 (E4)	X530 37; X522 4	CAVC_SUM_ DO	display only	V1
O:030/03	5	CAVD INTERLOCK SUM	Panel J3, WD-02 contact 24 (E4)	X530 49; X522 5	CAVD_SUM_ DO	display only	V1
O:030/04	6	KLYSTRON INTERLOCK SUM	Panel J3, WD-02 contact 11 (E4)	X530 36; X522 6	KLY_SUM_ DO	display only	V1
O:030/05	7	FILAMENT TIME OUT	Panel J2 10	X530 10; X522 7	FIL_TIME_ OUT_DO	display only	V1
O:030/06	8	FOCUS INTERLOCK SUM	Panel J3, WD-02 contact 10 (E4)	X530 35; X522 8	FOCUS_SUM_ DO	display only	V1
O:030/07	9	W/G LOAD #1 SUM *The panel calls it Magic Tee 1*	Panel J3, WD-02 contact 22 (E4)	X530 47; X522 9	WG_1_SUM_ DO	display only	V1

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
O:030/10	12	W/G #2 SUM *The panel calls it Magic Tee 2*	Panel J3, WD-02 contact 9 (E4)	X530 34; X522 12	WG_2_SUM_ DO	display only	V1
O:030/11	13	W/G #3 SUM *The panel calls it Magic Tee 3*	Panel J3, WD-02 contact 21 (E4)	X530 46; X522 13	WG_3_SUM_ DO	display only	V1
O:030/12	14	W/G AIR PRESSURE	Panel J3, WD-02 contact 8 (E4)	X530 33; X522 14	WG_4_SUM_ DO	P3	V1
O:030/13	15	CIRCULATOR INTERLOCK SUM	Panel J3, WD-02 contact 20 (E4)	X530 45; X522 15	CIRC_SUM_ DO	display only	V1
O:030/14	16	FLOW INTERLOCK SUM	Panel J3, WD-02 contact 7 (E4)	X530 32; X522 16	FLOW_SUM_ DO	display only	V1
O:030/15	17	VACUUM INTERLOCK SUM	Panel J3, WD-02 contact 19 (E4)	X530 44; X522 17	VACUUM_ SUM_DO	display only	V1
O:030/16	18	TEMPERATURE INTERLOCK SUM	Panel J3, WD-02 contact 6 (E4)	X530 31; X522 18	TEMP_SUM_ DO	display only	V1
O:030/17	19	Waveguide air MPS; drawn SPARE *E8: WD-340-330-25 stops at X522 19*	Panel J3, WD-02 contact 17 (E4)	X530 42; X522 19	VXI_SUM_DO	open	V1
O:031/00	22	GLOBAL INTERLOCK SUM *WD-340-330-25 draws X530 66, panel 40 (E6)*	Panel J3, WD-02 contact 5 (E4)	X530 30; X522 22	GLOBAL_DO	P5	V1
O:031/01	23	MAIN FOCUS ON/OFF *WD-340-330-25 draws X530 67, panel 41 (E6)*	Panel J2 6	X530 6; X522 23	FOCUS_1_DO	display only	V1
O:031/02	24	BUCKING COIL ON/OFF *WD-340-330-25 draws X530 68, panel 42 (E6)*	Panel J2 19	X530 19; X522 24	FOCUS_2_DO	display only	V1
O:031/03	25	FILAMENT ON/OFF *Filament chassis branch: WD-340-330-28; WD-340-330-25 draws X530 69, panel 43 (E6)*	Panel J2 9; filament chassis X610-J1 A	X530 9 and X525 21; X522 25	FIL_LOW_DO	display only	V1
O:031/04	26	FILAMENT CURRENT UP *WD-340-330-28; WD-340-330-25 draws X530 70, panel 44 (E6)*	Filament chassis X610-J1 C (motor control)	X525 23; X522 26	FIL_HIGH_ DO	–	V1
O:031/05	27	A/B HEARTBEAT *WD-340-330-25 draws X530 71, X330-J1 12 (E6)*	Panel J3, WD-02 contact 18 (E4)	X530 43; X522 27	HEARTBEAT	P6	V1
O:031/06	28	COLLECTOR INTERLOCK SUM *WD-340-330-25 draws X530 72, panel 45 (E6)*	Panel J3, WD-02 contact 23 (E4)	X530 48; X522 28	COLL_ PROTECTION	P4	V1

Address	Term.	Signal	Far end	Through	Symbol	Chassis	V
O:031/07	29	HV DETECTED *WD-340-330-25 draws X530 73, panel 46 (E6)*	Panel J2 11	X530 11; X522 29	HVPS_ON	display only	V1
O:031/10	32	RF DETECTED *WD-340-330-25 draws X530 74, panel 47 (E6)*	Panel J2 24	X530 24; X522 32	RF_ON	display only	V1
O:031/11	33	KLYSTRON WINDOW AIR BLOWER ON/OFF *On with the filament, off 10 min after it; WD-340-330-25 draws X530 75, panel 48 (E6)*	Blower switches (WD-340-330-33)	X512 11; X522 33	BLOWER	–	V1
O:031/12	34	MAIN FOCUS INTERLOCK *E8: WD-340-330-25 stops at X522 34*	Panel J2 7	X530 7; X522 34	FOCUS_SUM_ 1_DO	P1	V1
O:031/13	35	BUCKING COIL INTERLOCK *E8: WD-340-330-25 stops at X522 35*	Panel J2 20	X530 20; X522 35	FOCUS_SUM_ 2_DO	P2	V1
O:031/14–17	36–39	SPARE	Not connected	X522 36–39	–	–	V1
DO2: X420 slot 12 (drawn RACK X4 MODULE 12); O:032, O:033; WD-340-330-26							
O:032/00	2	Driven; drawn SPARE *E11: follows the shield-rack air-temperature latch (FC22)*	Not recorded	X523 2	Unnamed; rung DIGI_ OUT1:020	–	V0
O:032/01–07	3–9	SPARE	Not connected	X523 3–9	–	–	V1
O:032/10–17	12–19	SPARE	Not connected	X523 12–19	–	–	V1
O:033/00–07	22–29	SPARE	Not connected	X523 22–29	–	–	V1
O:033/10–17	32–39	SPARE	Not connected	X523 32–39	–	–	V1

6.6 Interfaces Not Tabulated in R0

The remaining interfaces of the legacy station are not tabulated pin by pin, each for the reason given; the field check named closes the gap.

- VXI Slot 5: no drawing of the MPS Shutoff module or of the wiring of the SPEAR MPS and orbit permits into it has been found (FC2). Its J3 output reaches the AIM through the SPARE FAULT input of the Fast Interlock Chassis (Table 11, pin 34).
- The arc-detector fibers of the Fast Interlock Chassis (receivers J17 to J28, transmitters J33 to J44): which detectors feed which channels is not recorded (FC5); the fibers move to the ARC4 at I1. The drawing WD-340-330-30 is for PEP-II.
- The detector of the Fast Interlock Chassis: its RF input J32 and the level and transfer curve of its output J31 (FC4); the PLC-5 end of that output is analog input 2, channel 10 (Table 16).
- The terminal strips TS-1 to TS-8 inside the HVPS controllers: the drawings disagree and a field survey is needed (FC9, FC10).
- The field side of the RF MPS sensors beyond the cross-connects (WD-340-330-03, -04, -28 and -31 to -34): their PLC-5 ends are in Tables 15 to 18, and the field wiring stays in place at I4.

- The tuner motor wiring, WD-340-330-27: chassis 340-315 to the motor drivers and the cavity junction boxes; it is replaced at I3 (Doc M), so it is not needed before then.
- The coax of the RF patch panel in B132-11 (link 12): SD-340-330-01 draws the four-cavity PEP-II plant, and the SPEAR3 jack map is FC8.
- The ARC4 modules, the Interface Chassis and its Waveform Buffer, and the interlocks of the two LLRF9 units: their designs are not released; they are added to §8 when they are.

7 Field Checks

Table 19 lists the 24 checks on the hardware that the documents still need, 23 of them open. Each replaces the open items named in its last column, which ask for the same check in different documents. The method is a proposal; each check is planned with the RF group, and where the station must be off, in a shutdown.

Table 19: Field checks.

ID	What to establish	How	Needed by	Open items
FC1	Which contact of the J3 cable plug carries the AB heartbeat: 21 (panel schematic) or 18 (WD-340-330-02). Settles the J3 numbering (E4).	Ask J. Sebek or J. Wachter; otherwise, station off, unplug J3 only and probe the plug (O5 note, section 5)	I2: chassis J3 wiring	Doc C O5 Doc M O2 Doc I O2
FC2	How the SPEAR MPS and orbit-interlock permits enter VXI Slot 5, the electrical form of its J3 output, and the current their sources can supply to a second input.	Trace and measure at Slot 5	I2: Interface Chassis installation	Doc L O13 Doc I O1 Doc C O7 Doc M O6
FC3	What the AIM does when its External MPS channel trips: whether it asserts RF_FAULT and drops the HVPS on request, as for an arc.	Test in the TUNE state: unplug the SPARE FAULT cable	I2: Interface Chassis installation	Doc I O3 Doc L O13
FC4	Levels and sense on J15 and on the 37-pin cable to the AIM; lengths of the reset and timeout-bypass pulses; RF level at J32 and the detector transfer curve.	Measure on the hardware or on the bench	I2: production design	Doc C O4 Doc M O3
FC5	Use of the AIM's other J15 outputs; which tunnel fibers feed the enabled arc channels; that the klystron-arc and transformer-arc triggers reach only the HVPS controller.	Trace fibers and cables. The rear-panel photograph of November 2026 shows receivers RX CH.1 and 2 (J17, J18) fibered and RX CH.3 to 12 capped, and transmitters TX CH.1, 2 and 4 to 11 fibered	I1: moving the arc fibers; I2	Doc M O3 Doc I O6
FC6	PLC-5 heartbeat period and duty cycle at the panel output and at the AIM input.	Oscilloscope	I2: watchdog design	Doc C O6 Doc M O4
FC7	B118 oscilloscope channels 3 and 4 (transformer current, SCR firing signal): pick-off points, amplitudes and cable route to B132.	Trace in the HVPS controller	I2: Waveform Buffer HVPS inputs	Doc C O8
FC8	Physical RF channel map: the 24 IQA inputs against connectors, cable labels and cavity identities.	Trace with the cable labels	RF cabling changes	Doc L O4

ID	What to establish	How	Needed by	Open items
FC9	Terminal strips TS-1 to TS-8 of the HVPS controller: complete identification, surveyed wiring separated from drawing-only assignments.	Survey	I7: controller rewiring	Doc L O3
FC10	TS-5 assignments: terminal and color disagreements between the controller and switchgear drawings.	Approved field trace	I7: contactor rewiring	Doc L O8
FC11	Manual ground switch: which NO or NC contact is landed and its sense (E9).	Field test	I7: grounding verification	Doc L O1
FC12	PPS pin pairs E–F, G–H, A–B and C–D, readback excitation and lamp states.	With the PPS group	I7: PPS approval	Doc L O5
FC13	PPS sequence and timing end to end: loss of either permit, stuck outputs, contactor blocking.	Test with the PPS group	I7: PPS approval	Doc L O10
FC14	EMERGENCY OFF end to end: which slot-6 inputs drop, latch and alarm, and whether the vacuum contactor opens.	Test	Before relying on the mushroom	Doc L O11
FC15	PS-3 5 V rail: does it still feed the gate-trigger BIAS inputs or is it grounded?	Check at the terminals	I7: controller refurbishment	Doc L O12
FC16	Crowbar configuration: SCR type and matching trigger-board revision for every stack.	Inspect	I7: board replacement	Doc L O7
FC17	Current-sense circuit: the 0.5 Ω burden and the Slot-9 input.	Inspect and measure	I7: regulator calibration	Doc L O9
FC18	PPS connector: the original Burndy circular 8-pin or its Souriau GOB1208PNE replacement.	Inspect	I7: PPS Interface Box fabrication	Doc 0 O2
FC19	Interlock input level set at the factory in the installed LLRF9 units (3.3, 5 or 24 V).	Inspect the units or ask the vendor	I2: board design	Doc C O2 Doc 0 O1
FC20	2021 LLRF9 setup: the LLRF Window Interlock Status chassis in rack B132-14 and how the 2021 tests were wired.	Inspect and document	I2: Interface Chassis installation	Doc M O7
FC21 *done*	Connections moved when the active HVPS unit is switched between SPEAR 1 and SPEAR 2.	Result: the switch-over procedure (J. Sebek, R0, January 2023) re-sets the bus bars of the cable switch tank, switches the fan breakers on panel HVPS2 in B514 (CB7-9 for SPEAR 2, CB8-10 for SPEAR 1) and, in the B118 junction box, re-plugs J1 to J6, the three fibers and the klystron-arc BNC to the unit in service. F. Wang confirmed on 12 November 2026 that the fibers are re-patched	I7: HVPS and PPS	Doc M O9
FC22	What output O:032/0 drives: output module 2, terminal X523-2, drawn spare on WD-340-330-26 but driven by the program from the shield-rack air-temperature latch (E11).	Trace from X523-2	I4: RF MPS	Doc M O8

ID	What to establish	How	Needed by	Open items
FC23	Catalog numbers of the thermocouple module (slot 3) and the input module (slot 7) of the PLC in each HVPS controller (E18).	Read the module labels: AB-1746, as on WD-730-790-02 and in the ladder program, or AB-1794, as on EI-444-636-00. Slot 3 has four channels: the drawings show IN0 to IN3, and the program writes four configuration words and reads eight input words, four data and four status	No step; the drawing record	none
FC24	The points of the RF MPS PLC-5 I/O list that the drawings leave open: the X519 terminal numbers (E7); the ion-pump supply pin of cavity B's pump input, drawn as pin 1 where the other cavities use pin 20; and whether anything is connected to analog input 1 channel 15, drawn NOT USED but read by EPICS as SRF1:KLYS:VACM.	Read the wire labels at X519, X520 and X525	I4: ControlLogix program and tags	none

8 The Upgraded Station

Table 20 lists the units of the upgrade with the integration step that brings each into service (Doc M); Table 3 gives the step that retires each legacy unit. A unit's place, connections and drawings are added here when its design is released, and the record of each step is complete when this document shows the station as it stands after it. The overview of the end state is drawn when the rack layout is fixed.

Table 20: Units of the upgrade.

Unit	Model or document	Status	In service
B132 (rack to be set)			
AFT ARC4 arc detection	AFT microwave ARC4	planned	from I1
Interface Chassis and Waveform Buffer	Doc C	planned	from I2
Galil tuner controller *driven by LLRF9 Unit 1*	DMC-4143	planned	from I3
RF MPS ControlLogix	Allen-Bradley 1756	planned	from I4
Klystron heater supply	TDK-Lambda Genesys	planned	from I5
IOC B132-IOCRF on new hardware	–	planned	from I8
B118			
New Hoffman box and PPS Interface Box per HVPS unit *first unit at I7b, second at I7c*	–	planned	from I7

A Data Files

The data are kept in `Designs/tex/docW-station-wiring/data/`, one file per table, in UTF-8 with a header row. A field that holds a comma is quoted; lists inside a field are separated by semicolons.

Table 21: Data files and their columns.

File	Feeds	Columns
`drawings.csv`	Tables 5 and 6	number, revision, title, area, status (Table 1), errata, files (repository paths; copies separated by semicolons), notes
`errata.csv`	Table 7	identifier, drawings, finding, evidence, status, note on the repository copy, documents where it is recorded
`checks.csv`	Table 19	identifier, what to establish, method, the step that needs it, the open items it replaces, status (open or done)
`devices.csv`	Tables 3 and 20	identifier, unit, model or drawing, location, place, status, the step that brings it in, the step that retires it, notes
`trunks.csv`	Table 4	number, from, to, medium, content, drawings
`connections.csv`	Tables 8 and 9	connector, contact, the other drawing's contact, labels of the two drawings, direction, X530 terminal, next point, PLC-5 point, use in the Interface Chassis, the step that retires it, verification level (Table 2)
`cables.csv`	Tables 10 to 13	cable, pin or strip, signal, the two ends, sense, use in the Interface Chassis, the step that retires it, verification level, notes
`plcmodules.csv`	Table 14	module, catalog number, location, PLC-5 address, drawing, content, notes
`plcio.csv`	Tables 15 to 18	module, channel or address, module terminal, signal, far end, cross-connects, EPICS record or program symbol, use in the Interface Chassis, verification level, notes

The numbers of Figure 1 are drawn by hand in `tikz/fig-station-overview.tex`; a change to `trunks.csv` needs the same change there.